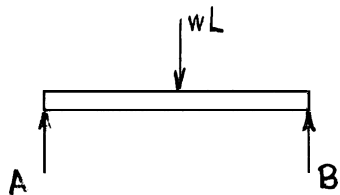


PROBLEM 5.1

For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

Reactions:



$$+\circlearrowleft \Sigma M_B = 0: -AL + wL \cdot \frac{L}{2} = 0 \quad A = \frac{wL}{2}$$

$$+\circlearrowleft \Sigma M_A = 0: BL - wL \cdot \frac{L}{2} = 0 \quad B = \frac{wL}{2}$$

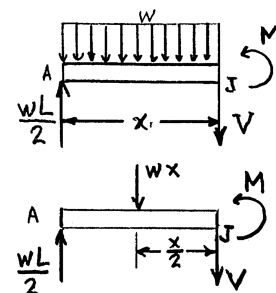
Free body diagram for determining reactions:

Over whole beam, $0 < x < L$

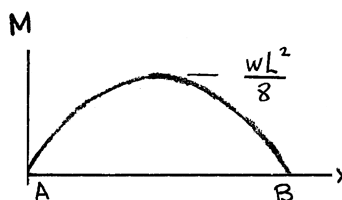
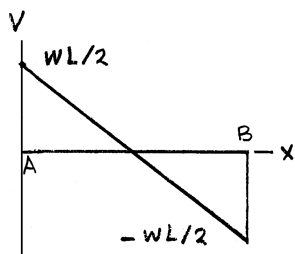
Place section at x .

Replace distributed load by equivalent concentrated load.

$$+\uparrow \Sigma F_y = 0: \frac{wL}{2} - wx - V = 0$$



$$V = w\left(\frac{L}{2} - x\right) \quad \blacktriangleleft$$



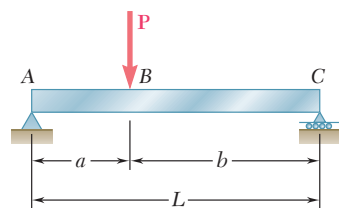
$$+\circlearrowleft \Sigma M_J = 0: -\frac{wL}{2}x + wx \cdot \frac{x}{2} + M = 0$$

$$M = \frac{w}{2}(Lx - x^2)$$

$$M = \frac{w}{2}x(L - x) \quad \blacktriangleleft$$

Maximum bending moment occurs at $x = \frac{L}{2}$.

$$M_{\max} = \frac{wL^2}{8} \quad \blacktriangleleft$$



PROBLEM 5.2

For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

Reactions:

$$(+\circlearrowleft \Sigma M_C = 0: LA - bP = 0 \quad A = \frac{Pb}{L}$$

$$(+\circlearrowleft \Sigma M_A = 0: LC - aP = 0 \quad C = \frac{Pa}{L}$$

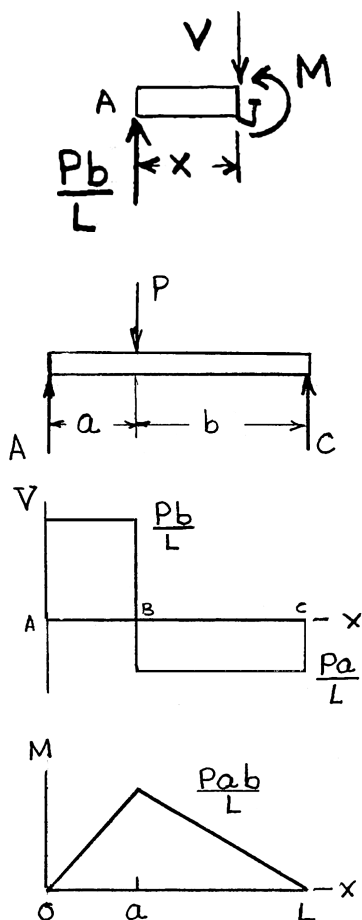
From A to B, $0 < x < a$

$$+\uparrow \Sigma F_y = 0: \frac{Pb}{L} - V = 0$$

$$V = \frac{Pb}{L} \quad \blacktriangleleft$$

$$(+\circlearrowleft \Sigma M_J = 0: M - \frac{Pb}{L}x = 0$$

$$M = \frac{Pbx}{L} \quad \blacktriangleleft$$



From B to C, $a < x < L$

$$+\uparrow \Sigma F_y = 0: V + \frac{Pa}{L} = 0$$

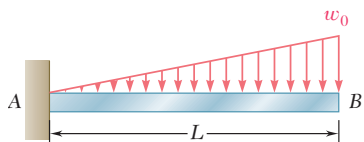
$$V = -\frac{Pa}{L} \quad \blacktriangleleft$$

$$(+\circlearrowleft \Sigma M_K = 0: -M + \frac{Pa}{L}(L - x) = 0$$

$$M = \frac{Pa(L - x)}{L} \quad \blacktriangleleft$$

At section B,

$$M = \frac{Pab}{L} \quad \blacktriangleleft$$

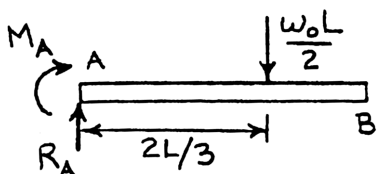


PROBLEM 5.3

For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

Free body diagram for determining reactions.



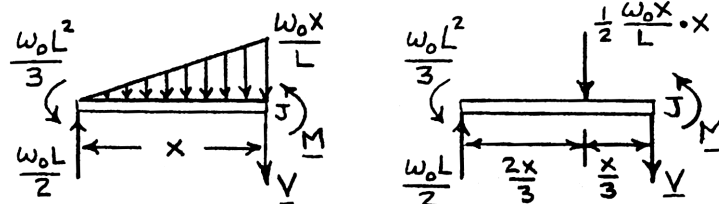
Reactions:

$$+\uparrow \Sigma F_y = 0: R_A - \frac{w_0 L}{2} = 0 \quad R_A = \frac{w_0 L}{2} \uparrow$$

$$+\curvearrowright \Sigma M_A = 0: -M_A - \left(\frac{w_0 L}{2} \right) \left(\frac{2L}{3} \right) = 0$$

$$M_A = -\frac{w_0 L^2}{3} = \frac{w_0 L^2}{3} \curvearrowleft$$

Use portion to left of the section as the free body.



Replace distributed load with equivalent concentrated load.

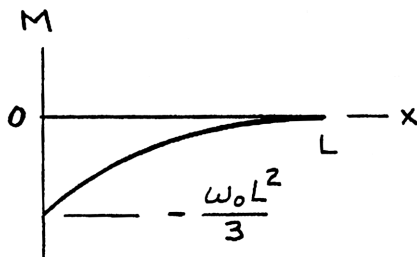
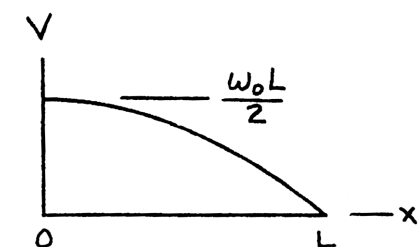
$$+\uparrow \Sigma F_y = 0: \frac{w_0 L}{2} - \frac{1}{2} \frac{w_0 x}{L} \cdot x - V = 0$$

$$V = \frac{w_0 L}{2} - \frac{w_0 x^2}{2L} \quad \blacktriangleleft$$

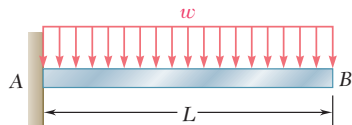
$$+\curvearrowright \Sigma M_J = 0:$$

$$\frac{w_0 L^2}{3} - \left(\frac{w_0 L}{2} \right) (x) + \left(\frac{1}{2} \frac{w_0 x}{L} \cdot x \right) \left(\frac{x}{3} \right) + M = 0$$

$$M = -\frac{w_0 L^2}{3} + \frac{w_0 L x}{2} - \frac{w_0 x^3}{6L} \quad \blacktriangleleft$$



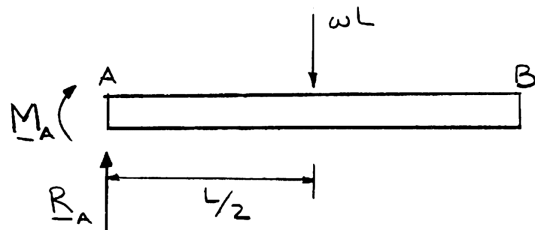
PROBLEM 5.4



For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

Free body diagram for determining reactions.



Reactions:

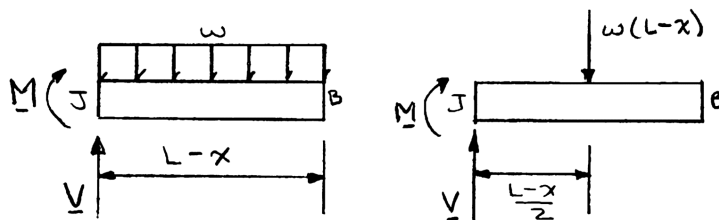
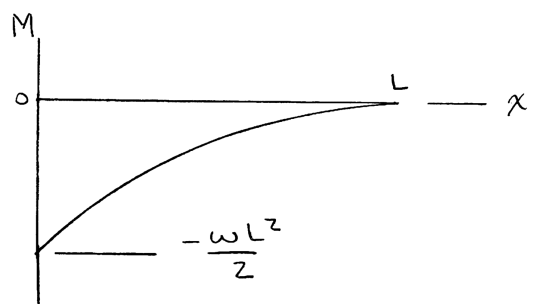
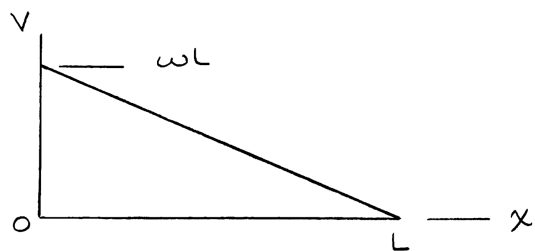
$$+\uparrow \Sigma F_y = 0: R_A - wL = 0$$

$$R_A = wL \uparrow$$

$$+\circlearrowleft \Sigma M_A = 0: -M_A - (wL)\left(\frac{L}{2}\right) = 0$$

$$M_A = -\frac{w_0 L^2}{2}$$

Use portion to the right of the section as the free body.



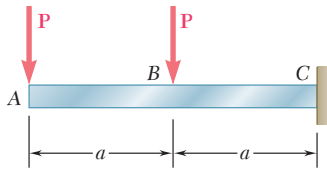
Replace distributed load by equivalent concentrated load.

$$+\uparrow \Sigma F_y = 0: V - w(L - x) = 0$$

$$V = w(L - x) \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: -M - w(L - x)\left(\frac{L - x}{2}\right) = 0$$

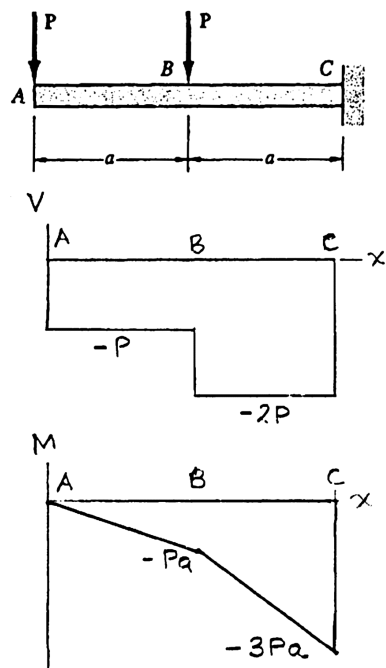
$$M = -\frac{w}{2}(L - x)^2 \quad \blacktriangleleft$$



PROBLEM 5.5

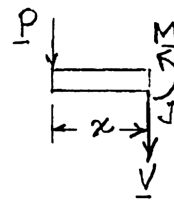
For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION



From A to B:

$$0 < x < a$$



$$+\uparrow \Sigma F_y = 0: \quad -P - V = 0$$

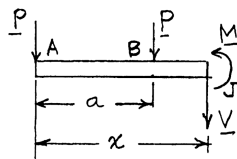
$$V = -P \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: \quad Px + M = 0$$

$$M = -Px \quad \blacktriangleleft$$

From B to C:

$$a < x \leq 2a$$

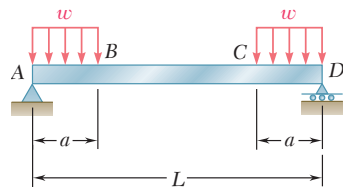


$$+\uparrow \Sigma F_y = 0: \quad -P - P - V = 0$$

$$V = -2P \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: \quad Px + P(x - a) + M = 0$$

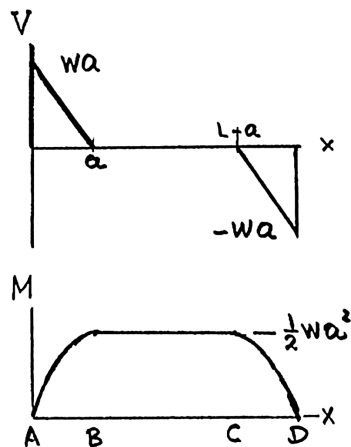
$$M = -2Px + Pa \quad \blacktriangleleft$$



PROBLEM 5.6

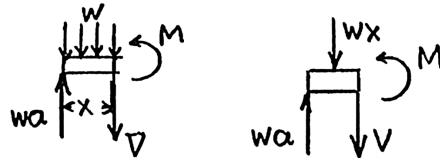
For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION



Reactions: $A = D = wa$

From A to B, $0 < x < a$



$$+\uparrow \Sigma F_y = 0: \quad wa - wx - V = 0$$

$$V = w(a - x) \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: \quad -wax + (wx)\frac{x}{2} + M = 0$$

$$M = w\left(ax - \frac{x^2}{2}\right) \quad \blacktriangleleft$$

From B to C, $a < x < L - a$

$$\Sigma F_y = 0: \quad wa - wa - V = 0$$

$$V = 0 \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: \quad -wax + wa\left(x - \frac{a}{2}\right) + M = 0 \quad M = \frac{1}{2}wa^2 \quad \blacktriangleleft$$

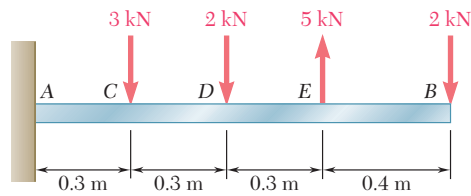
From C to D, $L - a < x < L$

$$+\uparrow \Sigma F_y = 0: \quad V - w(L - x) + wa = 0$$

$$V = w(L - x - a) \quad \blacktriangleleft$$

$$+\circlearrowleft \Sigma M_J = 0: \quad -M - w(L - x)\left(\frac{L - x}{2}\right) + wa(L - x) = 0$$

$$M = w\left[a(L - x) + \frac{1}{2}(L - x)^2\right] \quad \blacktriangleleft$$

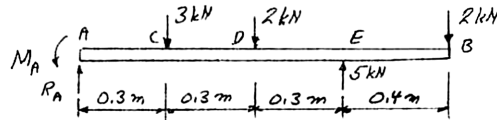


PROBLEM 5.7

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Origin at A:



Reaction at A:

$$\begin{aligned}\uparrow \Sigma F_y = 0: \quad R_A - 3 - 2 + 5 - 2 &= 0 \quad R_A = 2 \text{ kN} \uparrow \\ + \circlearrowleft \Sigma M_A = 0: \quad M_A - (3 \text{ kN})(0.3 \text{ m}) - (2 \text{ kN})(0.6 \text{ m}) + (5 \text{ kN})(0.9 \text{ m}) - (2 \text{ kN})(1.3 \text{ m}) &= 0 \\ M_A &= 0.2 \text{ kN} \cdot \text{m} \quad \circlearrowleft \blacktriangleleft\end{aligned}$$

From A to C:

$$\begin{aligned}\text{Free-body diagram (1):} \quad M_A = 0.2 \text{ kN} \cdot \text{m}, \quad R_A = 2 \text{ kN} \uparrow \\ \Sigma F_y = 0: \quad V = 2 \text{ kN} \\ + \circlearrowleft \Sigma M_1 = 0: \quad 0.2 \text{ kN} \cdot \text{m} - (2 \text{ kN})x + M = 0 \\ M = -0.2 + 2x\end{aligned}$$

From C to D:

$$\begin{aligned}\text{Free-body diagram (2):} \quad M_A = 0.2 \text{ kN} \cdot \text{m}, \quad R_A = 2 \text{ kN} \uparrow \\ \uparrow \Sigma F_y = 0: \quad 2 - 3 - V = 0 \quad V = -1 \text{ kN} \\ + \circlearrowleft \Sigma M_2 = 0: \quad + 0.2 \text{ kN} \cdot \text{m} - (2 \text{ kN})x + (3 \text{ kN})(x - 0.3) + M = 0 \\ M = 0.7 - x\end{aligned}$$

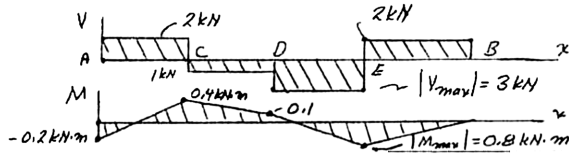
From D to E:

$$\begin{aligned}\text{Free-body diagram (3):} \quad \text{Shear force } V \text{ and bending moment } M \text{ at D.} \\ \Sigma F_y = 0: \quad V + 5 - 2 = 0 \quad V = -3 \text{ kN} \\ + \circlearrowleft \Sigma M_3 = 0: \quad -M + 5(0.9 - x) - (2)(1.3 - x) = 0 \\ M = +1.9 - 3x\end{aligned}$$

PROBLEM 5.7 (Continued)

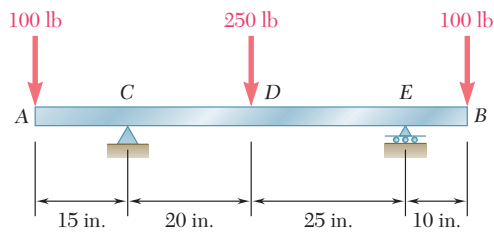
From *E* to *B*:

$$\begin{aligned} \sum F_y = 0: \quad V &= 2 \text{ kN} \\ + \sum M_A = 0: \quad -M - 2(1.3 - x) &= 0 \\ M &= -2.6 + 2x \end{aligned}$$



(a) $|V|_{\max} = 3.00 \text{ kN} \blacktriangleleft$

(b) $|M|_{\max} = 0.800 \text{ kN} \cdot \text{m} \blacktriangleleft$

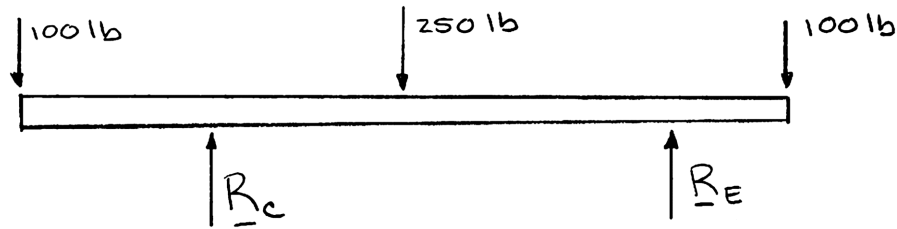


PROBLEM 5.8

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Reactions:



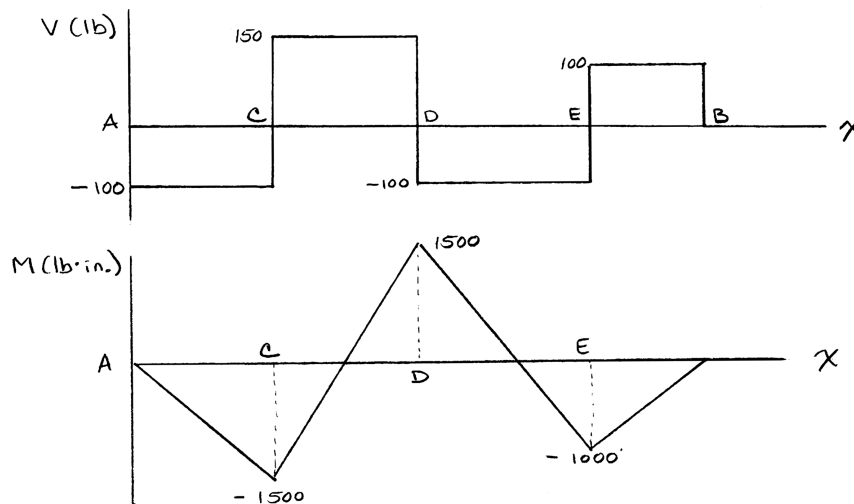
$$+\circlearrowleft \Sigma M_C = 0: \quad R_E(45 \text{ in.}) + 100 \text{ lb}(15 \text{ in.}) - 250 \text{ lb}(20 \text{ in.}) - 100 \text{ lb}(55 \text{ in.}) = 0$$

$$R_E = 200 \text{ lb}$$

$$+\uparrow \Sigma F_y = 0: \quad R_C + 200 \text{ lb} - 100 \text{ lb} - 250 \text{ lb} - 100 \text{ lb} = 0$$

$$R_C = 250 \text{ lb}$$

At any point, V is the sum of the loads and reactions to the left (assuming $+\uparrow$) and M the sum of their moments about that point (assuming $+\circlearrowleft$).



$$(a) \quad V_{\max} = 150.0 \text{ lb} \quad \blacktriangleleft$$

$$(b) \quad M_{\max} = 1500 \text{ lb} \cdot \text{in.} \quad \blacktriangleleft$$

PROBLEM 5.8 (Continued)

Detailed computations of moments:

$$M_A = 0$$

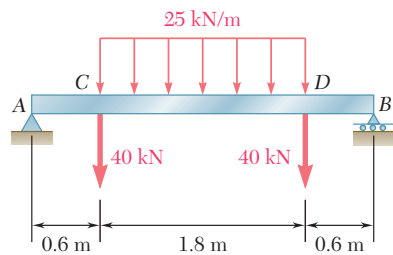
$$M_C = -(100 \text{ lb})(15 \text{ in.}) = -1500 \text{ lb} \cdot \text{in.}$$

$$M_D = -(100 \text{ lb})(35 \text{ in.}) + (250 \text{ lb})(20 \text{ in.}) = +1500 \text{ lb} \cdot \text{in.}$$

$$M_E = -(100 \text{ lb})(60 \text{ in.}) + (250 \text{ lb})(45 \text{ in.}) - (250 \text{ lb})(25 \text{ in.}) = -1000 \text{ lb} \cdot \text{in.}$$

$$M_B = -(100 \text{ lb})(70 \text{ in.}) + (250 \text{ lb})(55 \text{ in.}) - (250 \text{ lb})(35 \text{ in.}) + (200 \text{ lb})(10 \text{ in.}) = 0$$

(Checks)



PROBLEM 5.9

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

The distributed load is replaced with an equivalent concentrated load of 45 kN to compute the reactions.

$$(25 \text{ kN/m})(1.8 \text{ m}) = 45 \text{ kN}$$

$$+\circlearrowleft \Sigma M_A = 0: -(40 \text{ kN})(0.6 \text{ m}) - 45 \text{ kN}(1.5 \text{ m}) - 40 \text{ kN}(2.4 \text{ m}) + R_B(3.0 \text{ m}) = 0$$

$$R_B = 62.5 \text{ kN}$$

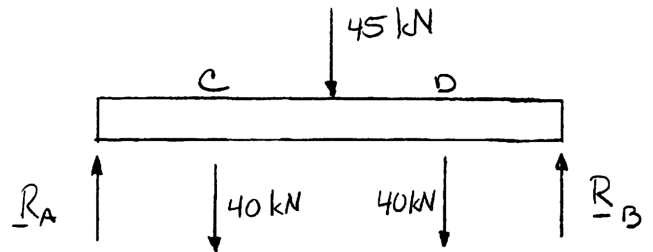
$$+\uparrow \Sigma F_y = 0: R_A + 62.5 \text{ kN} - 40 \text{ kN} - 45 \text{ kN} - 40 \text{ kN} = 0$$

$$R_A = 62.5 \text{ kN}$$

At C:

$$+\uparrow \Sigma F_y = 0: V = 62.5 \text{ kN}$$

$$+\circlearrowleft \Sigma M_1 = 0: M = (62.5 \text{ kN})(0.6 \text{ m}) = 37.5 \text{ kN} \cdot \text{m}$$



At centerline of the beam:

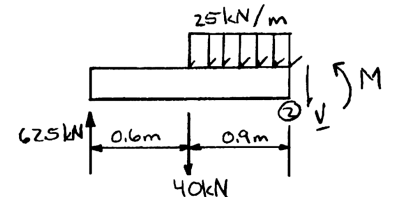
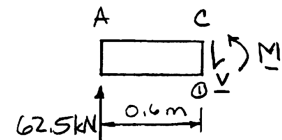
$$+\uparrow \Sigma F_y = 0: 62.5 \text{ kN} - 40 \text{ kN} - (25 \text{ kN/m})(0.9 \text{ m}) - V = 0$$

$$V = 0$$

$$+\circlearrowleft \Sigma M_2 = 0:$$

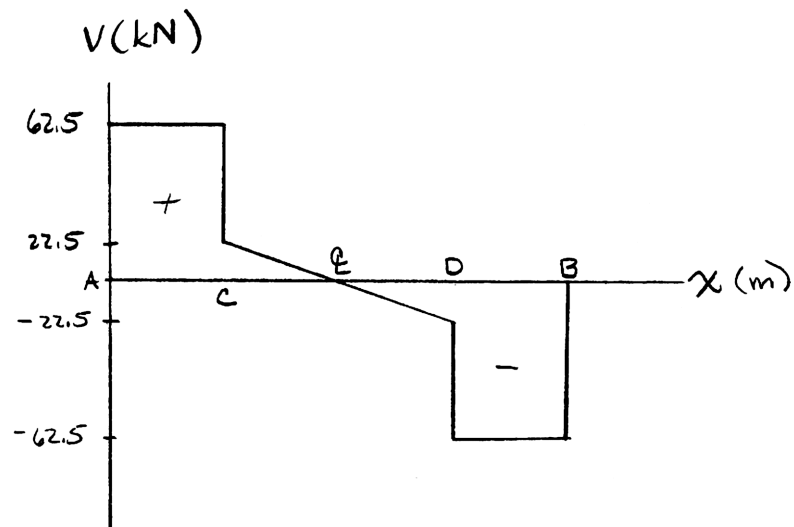
$$M - (62.5 \text{ kN})(1.5 \text{ m}) + (40 \text{ kN})(0.9 \text{ m}) + (25 \text{ kN/m})(0.9 \text{ m})(0.45 \text{ m}) = 0$$

$$M = 47.625 \text{ kN} \cdot \text{m}$$

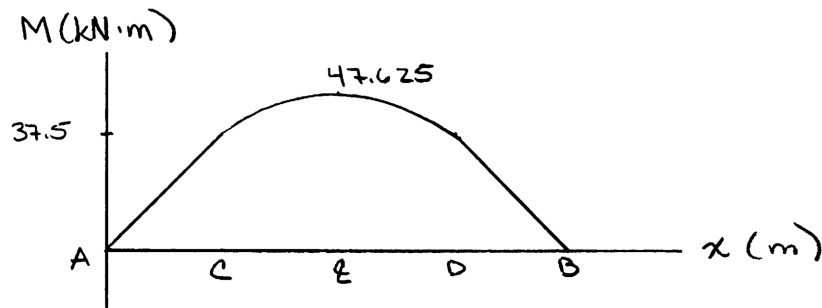


PROBLEM 5.9 (Continued)

Shear and bending-moment diagrams:



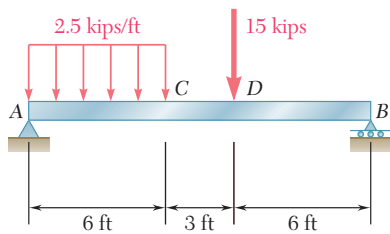
(a) $|V|_{\max} = 62.5 \text{ kN} \blacktriangleleft$



(b) $|M|_{\max} = 47.6 \text{ kN} \cdot \text{m} \blacktriangleleft$

From A to C and D to B, V is uniform; therefore M is linear.

From C to D, V is linear; therefore M is parabolic.



PROBLEM 5.10

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

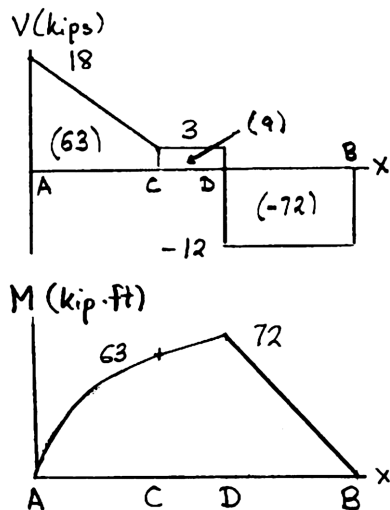
SOLUTION

$$+\circlearrowleft \Sigma M_B = 0: -15R_A + (12)(6)(2.5) + (6)(15) = 0$$

$$R_A = 18 \text{ kips}$$

$$+\circlearrowleft \Sigma M_A = 0: 15R_B - (3)(6)(2.5) - (9)(15) = 0$$

$$R_B = 12 \text{ kips}$$



Shear:

$$V_A = 18 \text{ kips}$$

$$V_C = 18 - (6)(2.5) = 3 \text{ kips}$$

$$C \text{ to } D: V = 3 \text{ kips}$$

$$D \text{ to } B: V = 3 - 15 = -12 \text{ kips}$$

Areas under shear diagram:

$$A \text{ to } C: \int V dx = \left(\frac{1}{2} \right) (6)(18 + 3) = 63 \text{ kip} \cdot \text{ft}$$

$$C \text{ to } D: \int V dx = (3)(3) = 9 \text{ kip} \cdot \text{ft}$$

$$D \text{ to } B: \int V dx = (6)(-12) = -72 \text{ kip} \cdot \text{ft}$$

Bending moments: $M_A = 0$

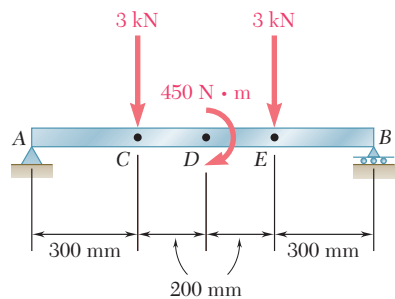
$$M_C = 0 + 63 = 63 \text{ kip} \cdot \text{ft}$$

$$M_D = 63 + 9 = 72 \text{ kip} \cdot \text{ft}$$

$$M_B = 72 - 72 = 0$$

$$|V|_{\max} = 18.00 \text{ kips} \quad \blacktriangleleft$$

$$|M|_{\max} = 72.0 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$



PROBLEM 5.11

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

$$+\circlearrowleft \sum M_B = 0: (700)(3) - 450 + (300)(3) - 1000A = 0$$

$$A = 2.55 \text{ kN} \uparrow$$

$$+\circlearrowleft \sum M_A = 0: -(300)(3) - 450 - (700)(3) + 1000B = 0$$

$$B = 3.45 \text{ kN} \uparrow$$

At A: $V = 2.55 \text{ kN}$ $M = 0$

A to C: $V = 2.55 \text{ kN}$

At C: $+\circlearrowleft \sum M_C = 0:$

$$-(300)(2.55) + M = 0$$

$$M = 765 \text{ N} \cdot \text{m}$$

C to E: $V = -0.45 \text{ N} \cdot \text{m}$

At D: $+\circlearrowleft \sum M_D = 0:$

$$-(500)(2.55) + (200)(3) + M = 0$$

$$M = 675 \text{ N} \cdot \text{m}$$

At D: $+\circlearrowleft \sum M_D = 0:$

$$-(500)(2.55) + (200)(3) - 450 + M = 0$$

$$M = 1125 \text{ N} \cdot \text{m}$$

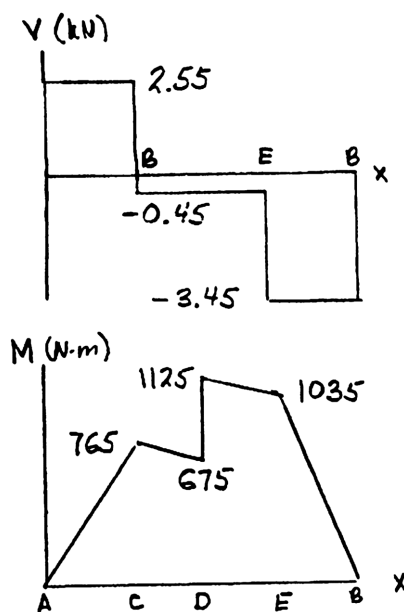
E to B: $V = -3.45 \text{ kN}$

At E: $+\circlearrowleft \sum M_E = 0:$

$$-M + (300)(3.45) = 0$$

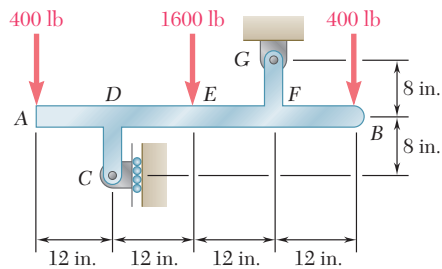
$$M = 1035 \text{ N} \cdot \text{m}$$

At B: $V = 3.45 \text{ kN}$, $M = 0$



$$(a) |V|_{\max} = 3.45 \text{ kN} \blacktriangleleft$$

$$(b) |M|_{\max} = 1125 \text{ N} \cdot \text{m} \blacktriangleleft$$

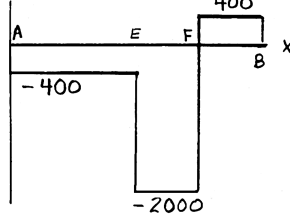


PROBLEM 5.12

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

V (lb.)



$$+\circlearrowleft \Sigma M_G = 0: -16C + (36)(400) + (12)(1600)$$

$$- (12)(400) = 0$$

$$C = 1800 \text{ lb}$$

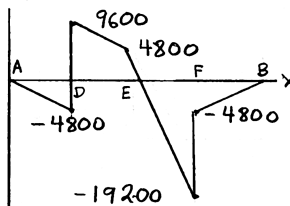
$$+\rightarrow \Sigma F_x = 0: -C + G_x = 0$$

$$G_x = 1800 \text{ lb}$$

$$+\uparrow \Sigma F_y = 0: -400 - 1600 + G_y - 400 = 0$$

$$G_y = 2400 \text{ lb}$$

M (lb·in)

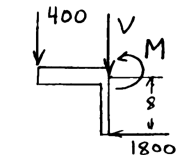
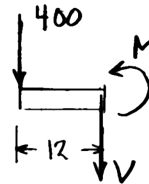


$$A \text{ to } E: V = -400 \text{ lb}$$

$$E \text{ to } F: V = -2000 \text{ lb}$$

$$F \text{ to } B: V = 400 \text{ lb}$$

$$\text{At } A \text{ and } B, M = 0$$



$$\text{At } D^-, +\circlearrowleft \Sigma M_D = 0: (12)(400) + M = 0$$

$$M = -4800 \text{ lb} \cdot \text{in.}$$

$$\text{At } D^+, +\circlearrowleft \Sigma M_D = 0: (12)(400) - (8)(1800) + M = 0$$

$$M = 9600 \text{ lb} \cdot \text{in.}$$

$$\text{At } E, +\circlearrowleft \Sigma M_E = 0: (24)(400) - (8)(1800) + M = 0$$

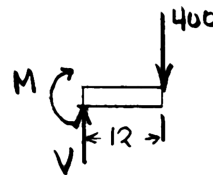
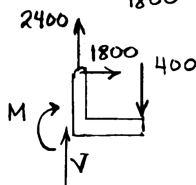
$$M = 4800 \text{ lb} \cdot \text{in.}$$

$$\text{At } F^-, +\circlearrowleft \Sigma M_F = 0: -M - (8)(1800) - (12)(400) = 0$$

$$M = -19,200 \text{ lb} \cdot \text{in.}$$

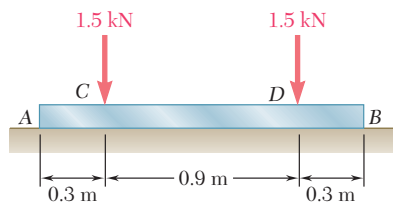
$$\text{At } F^+, +\circlearrowleft \Sigma M_F = 0: -M - (12)(400) = 0$$

$$M = -4800 \text{ lb} \cdot \text{in.}$$



$$(a) \quad \text{Maximum } |V| = 2000 \text{ lb} \quad \blacktriangleleft$$

$$(b) \quad \text{Maximum } |M| = 19,200 \text{ lb} \cdot \text{in.} \quad \blacktriangleleft$$



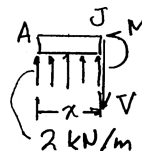
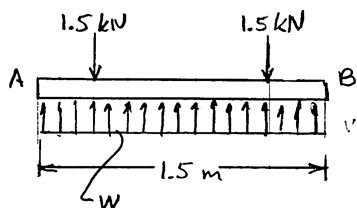
PROBLEM 5.13

Assuming that the reaction of the ground is uniformly distributed, draw the shear and bending-moment diagrams for the beam AB and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Over the whole beam,

$$+\uparrow \Sigma F_y = 0: 1.5w - 1.5 - 1.5 = 0 \quad w = 2 \text{ kN/m}$$



A to C: $0 \leq x < 0.3 \text{ m}$

$$+\uparrow \Sigma F_y = 0: 2x - V = 0 \quad V = (2x) \text{ kN}$$

$$+\circlearrowleft \Sigma M_J = 0: -(2x)\left(\frac{x}{2}\right) + M = 0 \quad M = (x^2) \text{ kN} \cdot \text{m}$$

At C^- , $x = 0.3 \text{ m}$

$$V = 0.6 \text{ kN}, \quad M = 0.090 \text{ kN} \cdot \text{m} \\ = 90 \text{ N} \cdot \text{m}$$

C to D: $0.3 \text{ m} < x < 1.2 \text{ m}$

$$+\uparrow \Sigma F_y = 0: 2x - 1.5 - V = 0 \quad V = (2x - 1.5) \text{ kN}$$

$$+\circlearrowleft \Sigma M_J = 0: -(2x)\left(\frac{x}{2}\right) + (1.5)(x - 0.3) + M = 0 \\ M = (x^2 - 1.5x + 0.45) \text{ kN} \cdot \text{m}$$

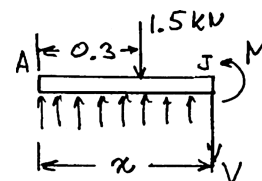
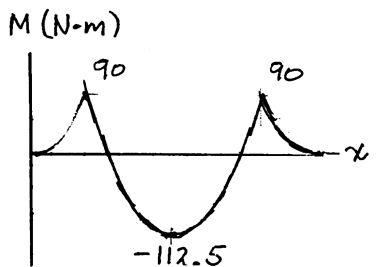
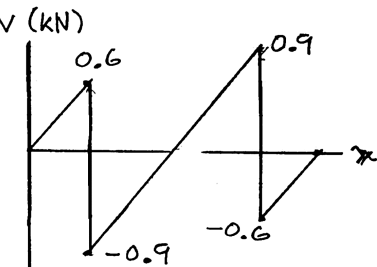
At the center of the beam, $x = 0.75 \text{ m}$

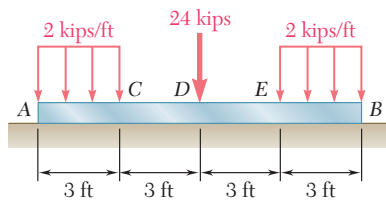
$$V = 0 \quad M = -0.1125 \text{ kN} \cdot \text{m} \\ = -112.5 \text{ N} \cdot \text{m}$$

At C^+ , $x = 0.3 \text{ m}$, $V = -0.9 \text{ kN}$

(a) Maximum $|V| = 0.9 \text{ kN} = 900 \text{ N}$ ◀

(b) Maximum $|M| = 112.5 \text{ N} \cdot \text{m}$ ◀

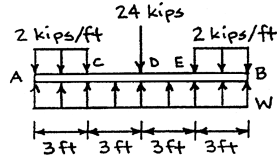




PROBLEM 5.14

Assuming that the reaction of the ground is uniformly distributed, draw the shear and bending-moment diagrams for the beam AB and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

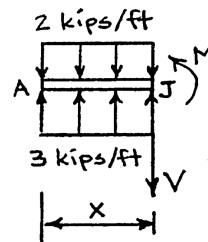
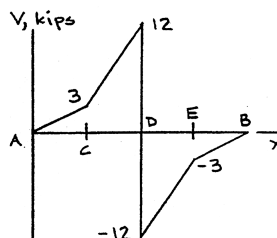
SOLUTION



Over the whole beam,

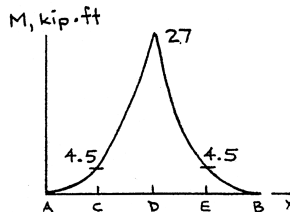
$$+\uparrow \Sigma F_y = 0: 12w - (3)(2) - 24 - (3)(2) = 0 \quad w = 3 \text{ kips/ft}$$

A to C: $(0 \leq x < 3 \text{ ft})$



$$+\uparrow \Sigma F_y = 0: 3x - 2x - V = 0 \quad V = (x) \text{ kips}$$

$$+\Sigma M_J = 0: -(3x)\frac{x}{2} + (2x)\frac{x}{2} + M = 0 \quad M = (0.5x^2) \text{ kip} \cdot \text{ft}$$



At C,

$$x = 3 \text{ ft}$$

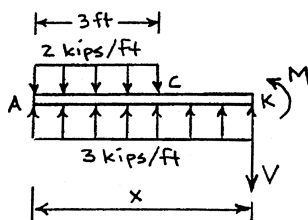
$$V = 3 \text{ kips}, \quad M = 4.5 \text{ kip} \cdot \text{ft}$$

C to D: $(3 \text{ ft} \leq x < 6 \text{ ft})$

$$+\uparrow \Sigma F_y = 0: 3x - (2)(3) - V = 0 \quad V = (3x - 6) \text{ kips}$$

$$+\Sigma M_K = 0: -(3x)\left(\frac{x}{2}\right) + (2)(3)\left(x - \frac{3}{2}\right) + M = 0$$

$$M = (1.5x^2 - 6x + 9) \text{ kip} \cdot \text{ft}$$



At D,

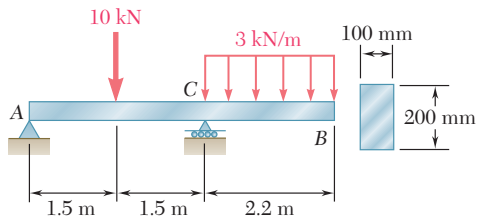
$$x = 6 \text{ ft}$$

$$V = 12 \text{ kips}, \quad M = 27 \text{ kip} \cdot \text{ft}$$

D to B: Use symmetry to evaluate.

$$(a) \quad |V|_{\max} = 12.00 \text{ kips} \quad \blacktriangleleft$$

$$(b) \quad |M|_{\max} = 27.0 \text{ kip} \cdot \text{ft} \quad \blacktriangleleft$$



PROBLEM 5.15

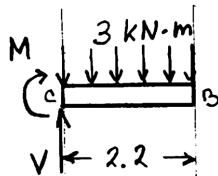
For the beam and loading shown, determine the maximum normal stress due to bending on a transverse section at C.

SOLUTION

Using *CB* as a free body,

$$+\circlearrowleft \Sigma M_C = 0: -M + (2.2)(3 \times 10^3)(1.1) = 0$$

$$M = 7.26 \times 10^3 \text{ N} \cdot \text{m}$$



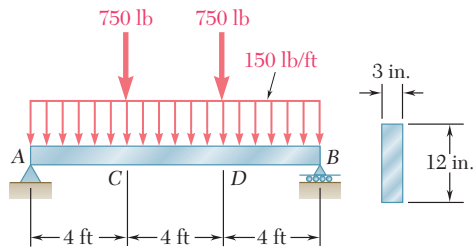
Section modulus for rectangle:

$$\begin{aligned} S &= \frac{1}{6}bh^2 \\ &= \frac{1}{6}(100)(200)^2 = 666.7 \times 10^3 \text{ mm}^3 \\ &= 666.7 \times 10^{-6} \text{ m}^3 \end{aligned}$$

Normal stress:

$$\sigma = \frac{M}{S} = \frac{7.26 \times 10^3}{666.7 \times 10^{-6}} = 10.8895 \times 10^6 \text{ Pa}$$

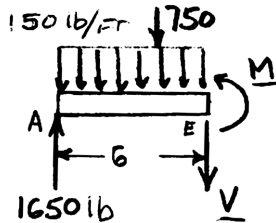
$$\sigma = 10.89 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.16

For the beam and loading shown, determine the maximum normal stress due to bending on a transverse section at C.

SOLUTION



Reactions: $C = A$ by symmetry.

$$+\uparrow \Sigma F_y = 0: \quad A + C - (2)(750) - (12)(150) = 0$$

$$A = C = 1650 \text{ lb}$$

Use left half of beam as free body.

$$+\curvearrowright \Sigma M_E = 0:$$

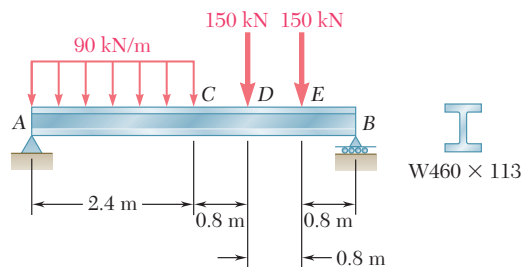
$$-(1650)(6) + (750)(2) + (150)(6)(3) + M = 0$$

$$M = 5700 \text{ lb} \cdot \text{ft} = 68.4 \times 10^3 \text{ lb} \cdot \text{in.}$$

Section modulus:
$$S = \frac{1}{6}bh^2 = \left(\frac{1}{6}\right)(3)(12)^2 = 72 \text{ in}^3$$

Normal stress:
$$\sigma = \frac{M}{S} = \frac{68.4 \times 10^3}{72} = 950 \text{ psi}$$

$$\sigma = 950 \text{ psi} \quad \blacktriangleleft$$

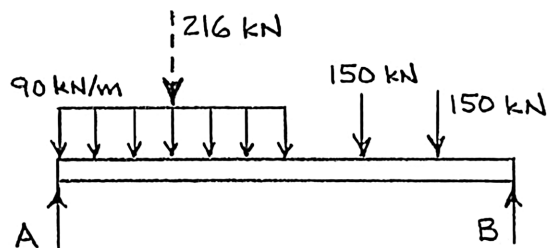


PROBLEM 5.17

For the beam and loading shown, determine the maximum normal stress due to bending on a transverse section at C.

SOLUTION

Use entire beam as free body.

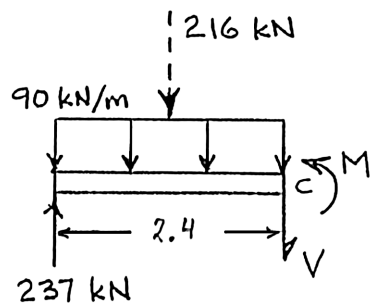


$$+\circlearrowleft \Sigma M_B = 0:$$

$$-4.8A + (3.6)(216) + (1.6)(150) + (0.8)(150) = 0$$

$$A = 237 \text{ kN} \uparrow$$

Use portion AC as free body.



$$+\circlearrowleft \Sigma M_C = 0:$$

$$M - (2.4)(237) + (1.2)(216) = 0$$

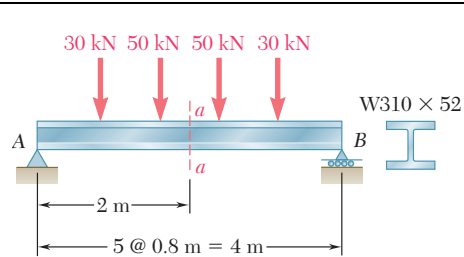
$$M = 309.6 \text{ kN} \cdot \text{m}$$

$$\text{For } W460 \times 113, S = 2390 \times 10^6 \text{ mm}^3$$

Normal stress:

$$\begin{aligned} \sigma &= \frac{M}{S} = \frac{309.6 \times 10^3 \text{ N} \cdot \text{m}}{2390 \times 10^{-6} \text{ m}^3} \\ &= 129.5 \times 10^6 \text{ Pa} \end{aligned}$$

$$\sigma = 129.5 \text{ MPa} \blacktriangleleft$$



PROBLEM 5.18

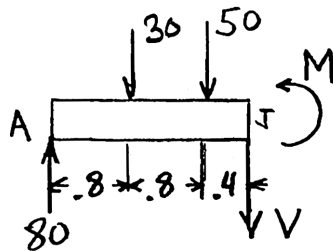
For the beam and loading shown, determine the maximum normal stress due to bending on section $a-a$.

SOLUTION

Reactions: By symmetry, $A = B$

$$+\uparrow \Sigma F_y = 0 : A = B = 80 \text{ kN} \uparrow$$

Using left half of beam as free body,



$$+\circlearrowleft \Sigma M_J = 0:$$

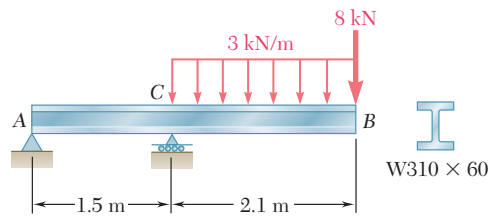
$$-(80)(2) + (30)(1.2) + (50)(0.4) + M = 0$$

$$M = 104 \text{ kN} \cdot \text{m} = 104 \times 10^3 \text{ N} \cdot \text{m}$$

For $W310 \times 52$, $S = 747 \times 10^3 \text{ mm}^3$
 $= 747 \times 10^{-6} \text{ m}^3$

Normal stress:
$$\sigma = \frac{M}{S} = \frac{104 \times 10^3}{747 \times 10^{-6}} = 139.2 \times 10^6 \text{ Pa}$$

$\sigma = 139.2 \text{ MPa} \quad \blacktriangleleft$



PROBLEM 5.19

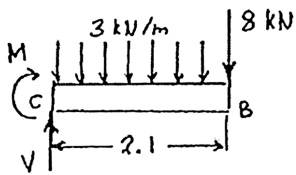
For the beam and loading shown, determine the maximum normal stress due to bending on a transverse section at C.

SOLUTION

Use portion CB as free body.

$$+\circlearrowleft \Sigma M_C = 0: \quad -M + (3)(2.1)(1.05) + (8)(2.1) = 0$$

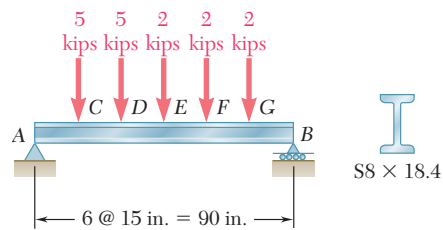
$$M = 23.415 \text{ kN} \cdot \text{m} = 23.415 \times 10^3 \text{ N} \cdot \text{m}$$



$$\begin{aligned} \text{For W310} \times 60, \quad S &= 844 \times 10^3 \text{ mm}^3 \\ &= 844 \times 10^{-6} \text{ m}^3 \end{aligned}$$

$$\text{Normal stress: } \sigma = \frac{|M|}{S} = \frac{23.415 \times 10^3}{844 \times 10^{-6}} = 27.7 \times 10^6 \text{ Pa}$$

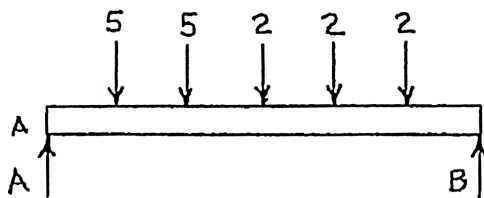
$$\sigma = 27.7 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.20

For the beam and loading shown, determine the maximum normal stress due to bending on a transverse section at C.

SOLUTION



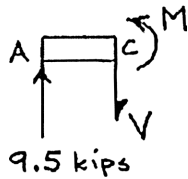
Use entire beam as free body.

$$+\circlearrowleft \sum M_B = 0:$$

$$-90A + (75)(5) + (60)(5) + (45)(2) + (30)(2) + (15)(2) = 0$$

$$A = 9.5 \text{ kips} \uparrow$$

Use portion AC as free body.



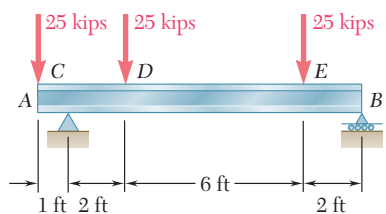
$$+\circlearrowleft \sum M_C = 0: \quad M - (15)(9.5) = 0$$

$$M = 142.5 \text{ kip} \cdot \text{in.}$$

For $S8 \times 18.4$, $S = 14.4 \text{ in}^3$

Normal stress:
$$\sigma = \frac{M}{S} = \frac{142.5}{14.4}$$

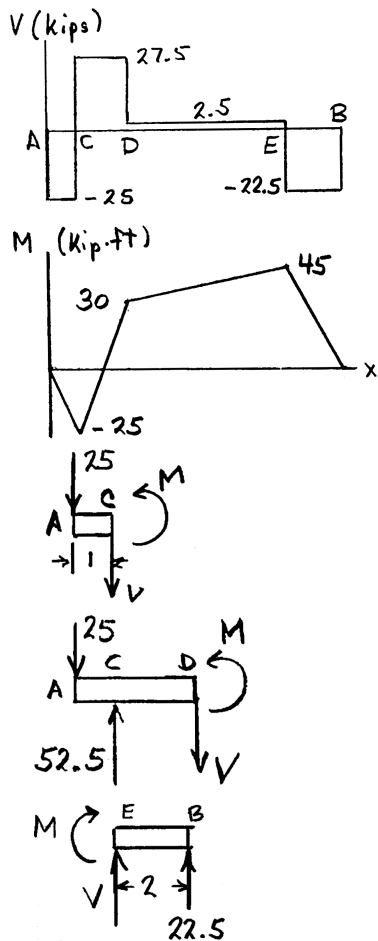
$$\sigma = 9.90 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 5.21

Draw the shear and bending-moment diagrams for the beam and loading shown and determine the maximum normal stress due to bending.

SOLUTION



$$+\circlearrowleft \sum M_B = 0:$$

$$(1)(25) - 10C + (8)(25) + (2)(25) = 0 \quad C = 52.5 \text{ kips} \uparrow$$

$$+\circlearrowleft \sum M_C = 0:$$

$$(1)(25) - (2)(25) - (8)(25) + 10B = 0 \quad B = 22.5 \text{ kips}$$

Shear:

$$A \text{ to } C^-: \quad V = -25 \text{ kips}$$

$$C^+ \text{ to } D^-: \quad V = 27.5 \text{ kips}$$

$$D^+ \text{ to } E^-: \quad V = 2.5 \text{ kips}$$

$$E^+ \text{ to } B: \quad V = -22.5 \text{ kips}$$

Bending moments:

$$\text{At } C, \quad +\circlearrowleft \sum M_C = 0: \quad (1)(25) + M = 0$$

$$M = -25 \text{ kip} \cdot \text{ft}$$

$$\text{At } D, \quad +\circlearrowleft \sum M_D = 0: \quad (3)(25) - (2)(52.5) + M = 0$$

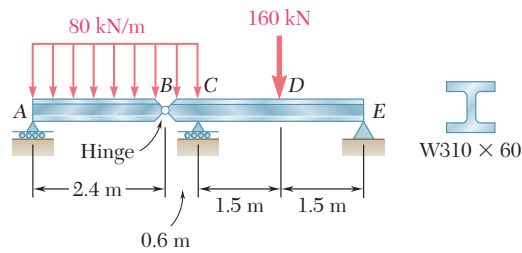
$$M = 30 \text{ kip} \cdot \text{ft}$$

$$\text{At } E, \quad +\circlearrowleft \sum M_E = 0: \quad -M + (2)(22.5) = 0 \quad M = 45 \text{ kip} \cdot \text{ft}$$

$$\max |M| = 45 \text{ kip} \cdot \text{ft} = 540 \text{ kip} \cdot \text{in.}$$

$$\text{For } S12 \times 35 \text{ rolled steel section, } S = 38.1 \text{ in}^3$$

$$\text{Normal stress: } \sigma = \frac{|M|}{S} = \frac{540}{38.1} = 14.17 \text{ ksi} \quad \sigma = 14.17 \text{ ksi} \quad \blacktriangleleft$$

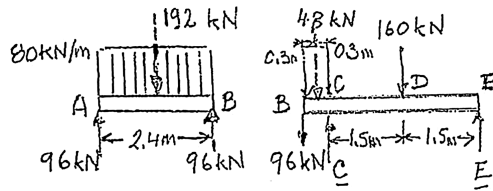


PROBLEM 5.22

Draw the shear and bending-moment diagrams for the beam and loading shown and determine the maximum normal stress due to bending.

SOLUTION

Statics: Consider portion AB and BE separately.



Portion BE :

$$+\circlearrowleft \sum M_E = 0:$$

$$(96)(3.6) + (48)(3.3) - C(3) + (160)(1.5) = 0$$

$$C = 248 \text{ kN } \uparrow$$

$$E = 56 \text{ kN } \uparrow$$

$$M_A = M_B = M_E = 0$$

At midpoint of AB :

$$\sum F_y = 0: \quad V = 0$$

$$\sum M = 0: \quad M = (96)(1.2) - (96)(0.6) = 57.6 \text{ kN} \cdot \text{m}$$

Just to the left of C :

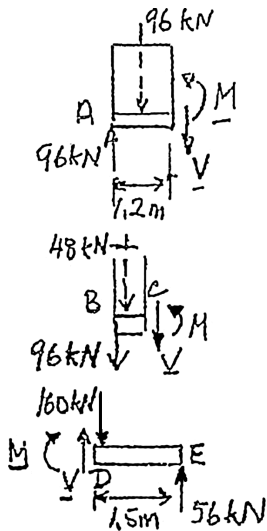
$$\sum F_y = 0: \quad V = -96 - 48 = -144 \text{ kN}$$

$$\sum M_C = 0: \quad M = -(96)(0.6) - (48)(0.3) = -72 \text{ kN} \cdot \text{m}$$

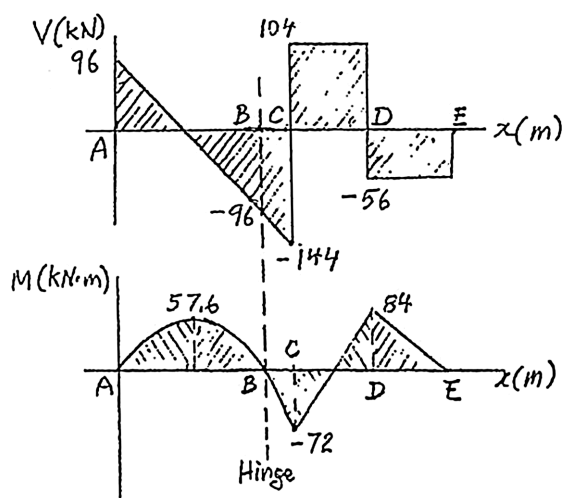
Just to the left of D :

$$\sum F_y = 0: \quad V = 160 - 56 = +104 \text{ kN}$$

$$\sum M_D = 0: \quad M = (56)(1.5) = +84 \text{ kN} \cdot \text{m}$$



PROBLEM 5.22 (Continued)



From the diagram,

$$|M|_{\max} = 84 \text{ kN} \cdot \text{m} = 84 \times 10^3 \text{ N} \cdot \text{m}$$

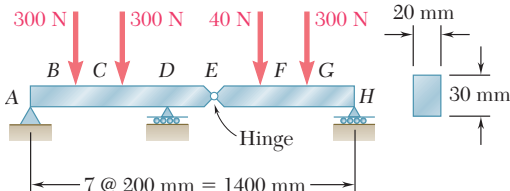
For W310 × 60 rolled-steel shape,

$$\begin{aligned} S_x &= 844 \times 10^3 \text{ mm}^3 \\ &= 844 \times 10^{-6} \text{ m}^3 \end{aligned}$$

$$\text{Stress: } \sigma_m = \frac{|M|_{\max}}{S}$$

$$\sigma_m = \frac{84 \times 10^3}{844 \times 10^{-6}} = 99.5 \times 10^6 \text{ Pa}$$

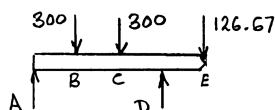
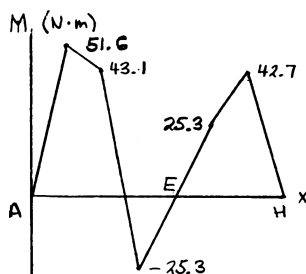
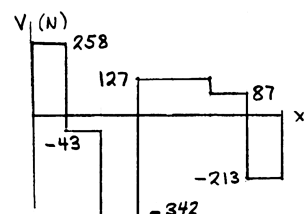
$$\sigma_m = 99.5 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.23

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum normal stress due to bending.

SOLUTION



Free body EFGH. Note that $M_E = 0$ due to hinge.

$$+\circlearrowleft \sum M_E = 0: 0.6 H - (0.2)(40) - (0.40)(300) = 0$$

$$H = 213.33 \text{ N}$$

$$+\uparrow \sum F_y = 0: V_E - 40 - 300 + 213.33 = 0$$

$$V_E = 126.67 \text{ N}$$

Shear:

$$E \text{ to } F: V = 126.67 \text{ N} \cdot \text{m}$$

$$F \text{ to } G: V = 86.67 \text{ N} \cdot \text{m}$$

$$G \text{ to } H: V = -213.33 \text{ N} \cdot \text{m}$$

Bending moment at F:

$$+\circlearrowleft \sum M_F = 0: M_F - (0.2)(126.67) = 0$$

$$M_F = 25.33 \text{ N} \cdot \text{m}$$

Bending moment at G:

$$+\circlearrowleft \sum M_G = 0: -M_G + (0.2)(213.33) = 0$$

$$M_G = 42.67 \text{ N} \cdot \text{m}$$

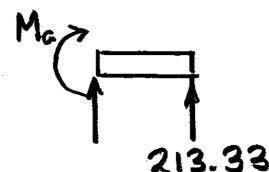
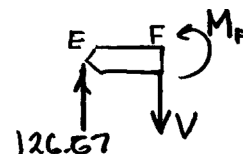
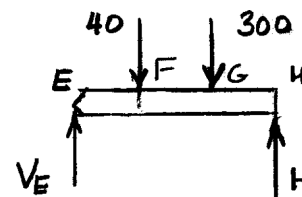
Free body ABCDE.

$$+\circlearrowleft \sum M_B = 0: 0.6 A + (0.4)(300) + (0.2)(300) - (0.2)(126.63) = 0$$

$$A = 257.78 \text{ N}$$

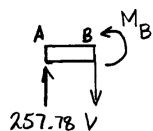
$$+\circlearrowleft \sum M_A = 0: -(0.2)(300) - (0.4)(300) - (0.8)(126.67) + 0.6 D = 0$$

$$D = 468.89 \text{ N}$$



PROBLEM 5.23 (Continued)

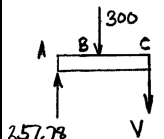
Bending moment at B.



$$+\circlearrowleft \Sigma M_B = 0: -(0.2)(257.78) + M_B = 0$$

$$M_B = 51.56 \text{ N} \cdot \text{m}$$

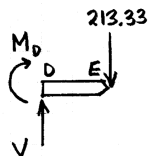
Bending moment at C.



$$+\circlearrowleft \Sigma M_C = 0: -(0.4)(257.78) + (0.2)(300) + M_C = 0$$

$$M_C = 43.11 \text{ N} \cdot \text{m}$$

Bending moment at D.



$$+\circlearrowleft \Sigma M_D = 0: -M_D - (0.2)(213.33) = 0$$

$$M_D = -25.33 \text{ N} \cdot \text{m}$$

$$\max |M| = 51.56 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

$$S = \frac{1}{6}bh^2 = \frac{1}{6}(20)(30)^2$$

$$= 3 \times 10^3 \text{ mm}^3 = 3 \times 10^{-6} \text{ m}^3$$

Normal stress:

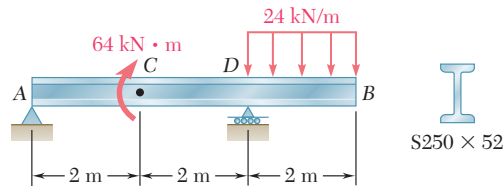
$$\sigma = \frac{51.56}{3 \times 10^{-6}} = 17.19 \times 10^6 \text{ Pa}$$

$$\sigma = 17.19 \text{ MPa} \quad \blacktriangleleft$$

$$|V|_{\max} = 342 \text{ N} \quad \blacktriangleleft$$

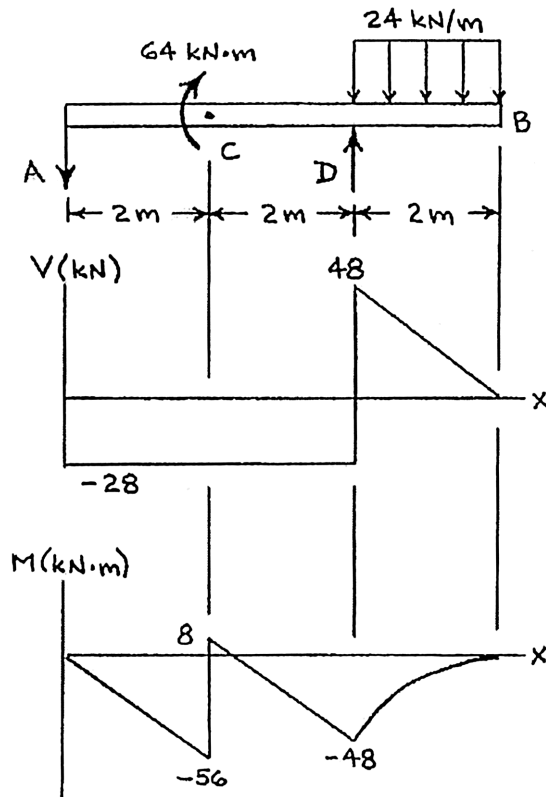
$$|M|_{\max} = 51.56 \text{ N} \cdot \text{m} \quad \blacktriangleleft$$

PROBLEM 5.24



Draw the shear and bending-moment diagrams for the beam and loading shown and determine the maximum normal stress due to bending.

SOLUTION



Reactions:

$$+\circlearrowleft \sum M_D = 0: 4A - 64 - (24)(2)(1) = 0 \quad A = 28 \text{ kN} \downarrow$$

$$+\uparrow \sum F_y = 0: -28 + D - (24)(2) = 0 \quad D = 76 \text{ kN} \uparrow$$

A to C: $0 < x < 2\text{ m}$

$$+\uparrow \sum F_y = 0: -V - 28 = 0 \quad V = -28 \text{ kN}$$

$$+\circlearrowleft \sum M_J = 0: M + 28x = 0 \quad M = (-28x) \text{ kN} \cdot \text{m}$$

C to D: $2\text{ m} < x < 4\text{ m}$

$$+\uparrow \sum F_y = 0: -V - 28 = 0 \quad V = -28 \text{ kN}$$

$$+\circlearrowleft \sum M_J = 0: M + 28x - 64 = 0 \quad M = (-28x + 64) \text{ kN} \cdot \text{m}$$

D to B: $4\text{ m} < x < 6\text{ m}$

$$+\uparrow \sum F_y = 0: V - 24(6 - x) = 0 \quad V = (-24x + 144) \text{ kN}$$

$$+\circlearrowleft \sum M_J = 0: -M - 24(6 - x)\left(\frac{6 - x}{2}\right) = 0$$

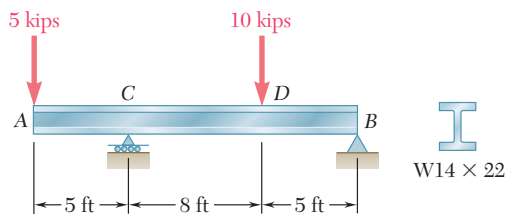
$$M = -12(6 - x)^2 \text{ kN} \cdot \text{m}$$

$$\max |M| = 56 \text{ kN} \cdot \text{m} = 56 \times 10^3 \text{ N} \cdot \text{m}$$

$$\text{For S250} \times 52 \text{ section, } S = 482 \times 10^3 \text{ mm}^3$$

$$\text{Normal stress: } \sigma = \frac{|M|}{S} = \frac{56 \times 10^3 \text{ N} \cdot \text{m}}{482 \times 10^{-6} \text{ m}^3} = 116.2 \times 10^6 \text{ Pa}$$

$$\sigma = 116.2 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.25

Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum normal stress due to bending.

SOLUTION

Reaction at C:
$$+\circlearrowleft \Sigma M_B = 0: (18)(5) - 13C + (5)(10) = 0$$

$$C = 10.769 \text{ kips}$$

Reaction at B:
$$+\circlearrowleft M_C = 0: (5)(5) - (8)(10) + 13B = 0$$

$$B = 4.231 \text{ kips}$$

Shear diagram:

A to C⁻: $V = -5 \text{ kips}$

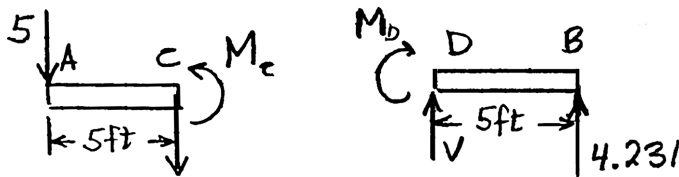
C⁺ to D⁻: $V = -5 + 10.769 = 5.769 \text{ kips}$

D⁺ to B: $V = 5.769 - 10 = -4.231 \text{ kips}$

At A and B, $M = 0$

At C,
$$+\circlearrowleft M_C = 0: (5)(5) + M_C = 0$$

$$M_C = -25 \text{ kip} \cdot \text{ft}$$



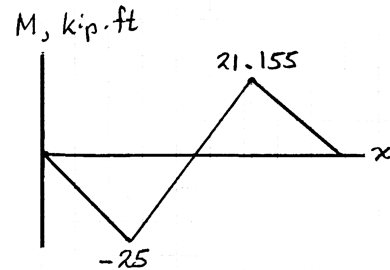
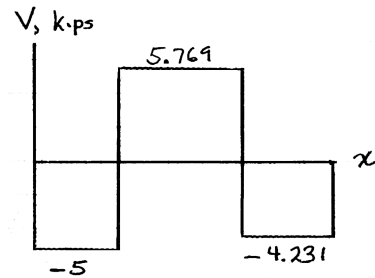
At D,
$$+\circlearrowleft \Sigma M_D = 0: -M_D + (5)(4.231)$$

$$M_D = 21.155 \text{ kip} \cdot \text{ft}$$

$|M|_{\max}$ occurs at C.

For W14 x 22 rolled-steel section, $S = 29.0 \text{ in}^3$

Normal stress:
$$\sigma = \frac{M}{S} = \frac{300}{29.0}$$



$|V|_{\max} = 5.77 \text{ kips} \blacktriangleleft$

$|M|_{\max} = 25 \text{ kip} \cdot \text{ft} = 300 \text{ kip} \cdot \text{in.} \blacktriangleleft$

$\sigma = 10.34 \text{ ksi} \blacktriangleleft$

PROBLEM 5.26

Knowing that $W = 12 \text{ kN}$, draw the shear and bending-moment diagrams for beam AB and determine the maximum normal stress due to bending.

SOLUTION

By symmetry, $A = B$

$$+\uparrow \Sigma F_y = 0: A - 8 + 12 - 8 + B = 0$$

$$A = B = 2 \text{ kN}$$

Shear:

A to C^- : $V = 2 \text{ kN}$

C^+ to D^- : $V = -6 \text{ kN}$

D^+ to E^- : $V = 6 \text{ kN}$

E^+ to B : $V = -2 \text{ kN}$

Bending moment:

At C , $+\circlearrowright \Sigma M_C = 0: M_C - (1)(2) = 0$

$$M_C = 2 \text{ kN} \cdot \text{m}$$

At D , $+\circlearrowright \Sigma M_D = 0: M_D - (2)(2) + (8)(1) = 0$

$$M_D = -4 \text{ kN} \cdot \text{m}$$

By symmetry, $M = 2 \text{ kN} \cdot \text{m}$ at D .

$$M_E = 2 \text{ kN} \cdot \text{m}$$

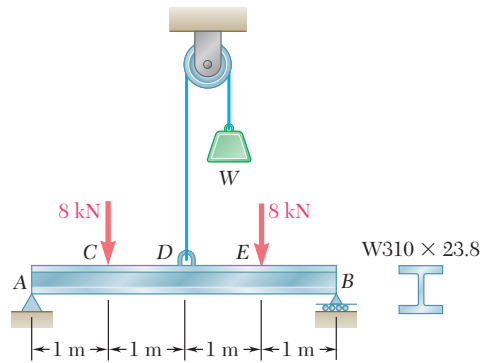
$\max |M| = 4.00 \text{ kN} \cdot \text{m}$ occurs at E .

For $W310 \times 23.8$, $S_x = 280 \times 10^3 \text{ mm}^3 = 280 \times 10^{-6} \text{ m}^3$

Normal stress:

$$\sigma_{\max} = \frac{|M|_{\max}}{S_x} = \frac{4 \times 10^3}{280 \times 10^{-6}}$$

$$= 14.29 \times 10^6 \text{ Pa} \quad \sigma_{\max} = 14.29 \text{ MPa}$$



PROBLEM 5.27

Determine (a) the magnitude of the counterweight W for which the maximum absolute value of the bending moment in the beam is as small as possible, (b) the corresponding maximum normal stress due to bending. (*Hint*: Draw the bending-moment diagram and equate the absolute values of the largest positive and negative bending moments obtained.)

SOLUTION

By symmetry, $A = B$

$$+\uparrow \Sigma F_y = 0: A - 8 + W - 8 + B = 0$$

$$A = B = 8 - 0.5W$$

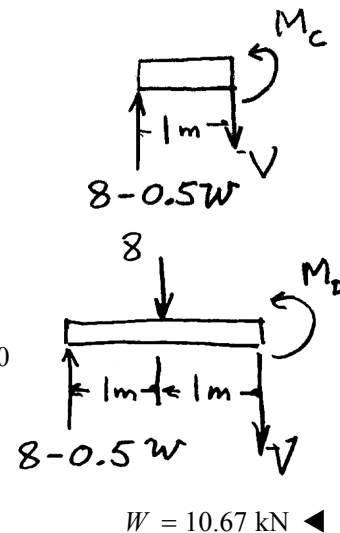
Bending moment at C: $+\circlearrowleft \Sigma M_C = 0: -(8 - 0.5W)(1) + M_C = 0$

$$M_C = (8 - 0.5W) \text{ kN} \cdot \text{m}$$

Bending moment at D: $+\circlearrowleft \Sigma M_D = 0: -(8 - 0.5W)(2) + (8)(1) + M_D = 0$

$$M_D = (8 - W) \text{ kN} \cdot \text{m}$$

Equate: $-M_D = M_C \quad W - 8 = 8 - 0.5W$



$$W = 10.67 \text{ kN} \quad \blacktriangleleft$$

(a) $W = 10.6667 \text{ kN}$

$$M_C = -2.6667 \text{ kN} \cdot \text{m}$$

$$M_D = 2.6667 \text{ kN} \cdot \text{m} = 2.6667 \cdot 10^3 \text{ N} \cdot \text{m}$$

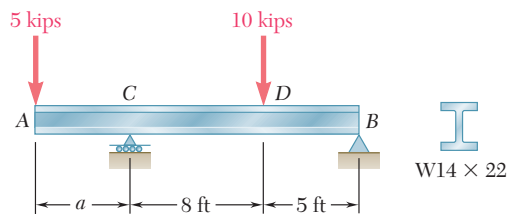
$$|M|_{\max} = 2.6667 \text{ kN} \cdot \text{m}$$

For W310 x 23.8 rolled-steel shape,

$$S_x = 280 \times 10^3 \text{ mm}^3 = 280 \times 10^{-6} \text{ m}^3$$

(b) $\sigma_{\max} = \frac{|M|_{\max}}{S_x} = \frac{2.6667 \times 10^3}{280 \times 10^{-6}} = 9.52 \times 10^6 \text{ Pa}$

$$\sigma_{\max} = 9.52 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.28

Determine (a) the distance a for which the maximum absolute value of the bending moment in the beam is as small as possible, (b) the corresponding maximum normal stress due to bending. (See hint of Prob. 5.27.)

SOLUTION

Reaction at B:

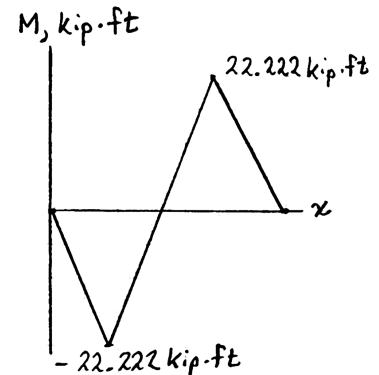
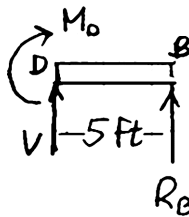
$$+\circlearrowleft \Sigma M_C = 0: 5a - (8)(10) + 13R_B = 0$$

$$R_B = \frac{1}{13}(80 - 5a)$$

Bending moment at D:

$$+\circlearrowleft \Sigma M_D = 0: -M_D + 5R_B = 0$$

$$M_D = 5R_B = \frac{5}{13}(80 - 5a)$$



Bending moment at C:

$$+\circlearrowleft M_C = 0 \quad 5a + M_C = 0$$

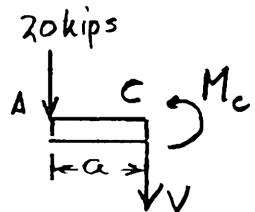
$$M_C = -5a$$

Equate:

$$-M_C = M_D$$

$$5a = \frac{5}{13}(80 - 5a)$$

$$a = 4.4444 \text{ ft}$$



$$(a) \quad a = 4.44 \text{ ft} \quad \blacktriangleleft$$

Then

$$-M_C = M_D = (5)(4.4444) = 22.222 \text{ kip} \cdot \text{ft}$$

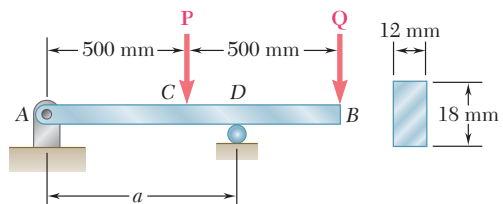
$$|M|_{\max} = 22.222 \text{ kip} \cdot \text{ft} = 266.67 \text{ kip} \cdot \text{in.}$$

For W14 x 22 rolled-steel section, $S = 29.0 \text{ in}^3$

Normal stress:

$$\sigma = \frac{M}{S} = \frac{266.67}{29.0} = 9.20 \text{ ksi}$$

$$(b) \quad 9.20 \text{ ksi} \quad \blacktriangleleft$$



PROBLEM 5.29

Knowing that $P = Q = 480$ N, determine (a) the distance a for which the absolute value of the bending moment in the beam is as small as possible, (b) the corresponding maximum normal stress due to bending. (See hint of Prob. 5.27.)

SOLUTION

$$P = 480 \text{ N} \quad Q = 480 \text{ N}$$

Reaction at A:

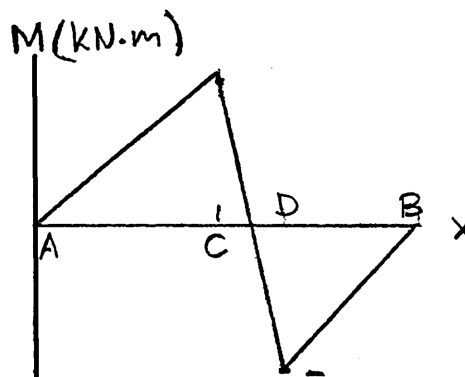
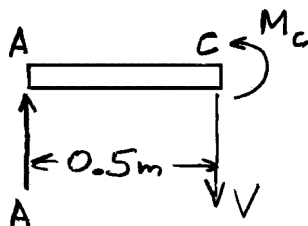
$$+\circlearrowleft \Sigma M_D = 0: -Aa + 480(a - 0.5) - 480(1 - a) = 0$$

$$A = \left(960 - \frac{720}{a} \right) \text{ N}$$

Bending moment at C:

$$+\circlearrowleft \Sigma M_C = 0: -0.5A + M_C = 0$$

$$M_C = 0.5A = \left(480 - \frac{360}{a} \right) \text{ N} \cdot \text{m}$$



Bending moment at D:

$$+\circlearrowleft \Sigma M_D = 0: -M_D - 480(1 - a) = 0$$

$$M_D = -480(1 - a) \text{ N} \cdot \text{m}$$

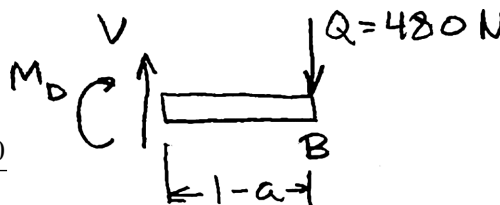
(a) Equate:

$$-M_D = M_C \quad 480(1 - a) = 480 - \frac{360}{a}$$

$$a = 0.86603 \text{ m}$$

$$a = 866 \text{ mm} \quad \blacktriangleleft$$

$$A = 128.62 \text{ N} \quad M_C = 64.31 \text{ N} \cdot \text{m} \quad M_D = -64.31 \text{ N} \cdot \text{m}$$



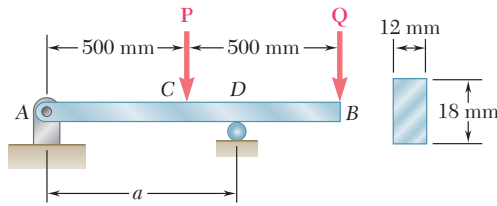
(b) For rectangular section, $S = \frac{1}{6}bh^2$

$$S = \frac{1}{6}(12)(13)^2 = 648 \text{ mm}^3 = 648 \times 10^{-9} \text{ m}^3$$

$$\sigma_{\max} = \frac{|M|_{\max}}{S} = \frac{64.31}{6.48 \times 10^{-9}} = 99.2 \times 10^6 \text{ Pa}$$

$$\sigma_{\max} = 99.2 \text{ MPa} \quad \blacktriangleleft$$

PROBLEM 5.30



Solve Prob. 5.29, assuming that $P = 480 \text{ N}$ and $Q = 320 \text{ N}$.

PROBLEM 5.29 Knowing that $P = Q = 480 \text{ N}$, determine (a) the distance a for which the absolute value of the bending moment in the beam is as small as possible, (b) the corresponding maximum normal stress due to bending. (See hint of Prob. 5.27.)

SOLUTION

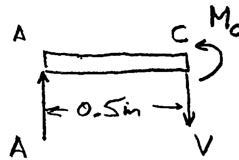
$$P = 480 \text{ N} \quad Q = 320 \text{ N}$$

$$\text{Reaction at A: } +\curvearrowright \Sigma M_D = 0: Aa + 480(a - 0.5) - 320(1 - a) = 0$$

$$A = \left(800 - \frac{560}{a} \right) \text{ N}$$

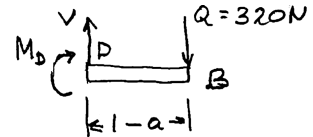
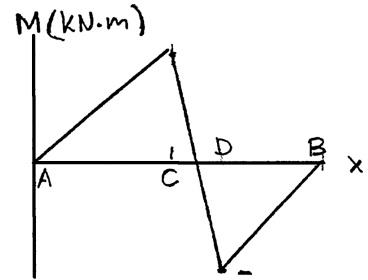
$$\text{Bending moment at C: } +\curvearrowright \Sigma M_C = 0: -0.5A + M_C = 0$$

$$M_C = 0.5A = \left(400 - \frac{280}{a} \right) \text{ N} \cdot \text{m}$$



$$\text{Bending moment at D: } +\curvearrowright \Sigma M_D = 0: -M_D - 320(1 - a) = 0$$

$$M_D = (-320 + 320a) \text{ N} \cdot \text{m}$$



$$\begin{aligned} \text{(a) Equate:} \quad -M_D &= M_C & 320 - 320a &= 400 - \frac{280}{a} \\ 320a^2 + 80a - 280 &= 0 & a &= 0.81873 \text{ m, } -1.06873 \text{ m} \end{aligned}$$

Reject negative root.

$$a = 819 \text{ mm} \quad \blacktriangleleft$$

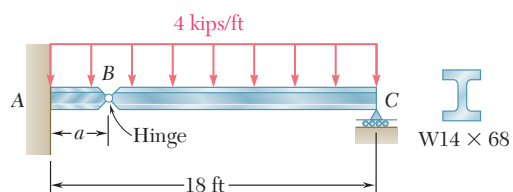
$$A = 116.014 \text{ N} \quad M_C = 58.007 \text{ N} \cdot \text{m} \quad M_D = -58.006 \text{ N} \cdot \text{m}$$

$$\text{(b) For rectangular section, } S = \frac{1}{6}bh^2$$

$$S = \frac{1}{6}(12)(18)^2 = 648 \text{ mm}^3 = 648 \times 10^{-9} \text{ m}^3$$

$$\sigma_{\max} = \frac{|M|_{\max}}{S} = \frac{58.0065}{648 \times 10^{-9}} = 89.5 \times 10^6 \text{ Pa}$$

$$\sigma_{\max} = 89.5 \text{ MPa} \quad \blacktriangleleft$$



PROBLEM 5.31

Determine (a) the distance a for which the absolute value of the bending moment in the beam is as small as possible, (b) the corresponding maximum normal stress due to bending. (See hint of Prob. 5.27.)

SOLUTION

For W14 × 68, $S_x = 103 \text{ in}^3$

Let $b = (18 - a) \text{ ft}$

Segment BC:

By symmetry, $V_B = C$

$$+\uparrow \Sigma F_y = 0: V_B + C - 4b = 0$$

$$V_B = 2b$$

$$+\circlearrowleft \Sigma M_J = 0: -V_B x + (4x)\left(\frac{x}{2}\right) - M = 0$$

$$M = V_B x - 2x^2 = 2bx - 2x^2 \text{ lb} \cdot \text{ft}$$

$$\frac{dM}{dx} = 2b - 2x_m = 0 \quad x_m = \frac{1}{2}b$$

$$M_{\max} = b^2 - \frac{1}{2}b^2 = \frac{1}{2}b^2$$

Segment AB:

$$+\circlearrowleft \Sigma M_K = 0: -4(a-x)\frac{(a-x)}{2}$$

$$-V_B(a-x) - M = 0$$

$$M = -2(a-x)^2 + 2b(a-x)$$

$|M_{\max}|$ occurs at $x = 0$.

$$|M_{\max}| = -2a^2 - 2ab = -2a^2 - 2a(18-a) = 36a$$

(a) Equate the two values of $|M_{\max}|$: $36a = \frac{1}{2}b^2 = \frac{1}{2}(18-a)^2 = 162 - 18a + \frac{1}{2}a^2$

$$\frac{1}{2}a^2 - 54a + 162 = 0 \quad a = 54 \pm \sqrt{(54)^2 - (4)\left(\frac{1}{2}\right)(162)}$$

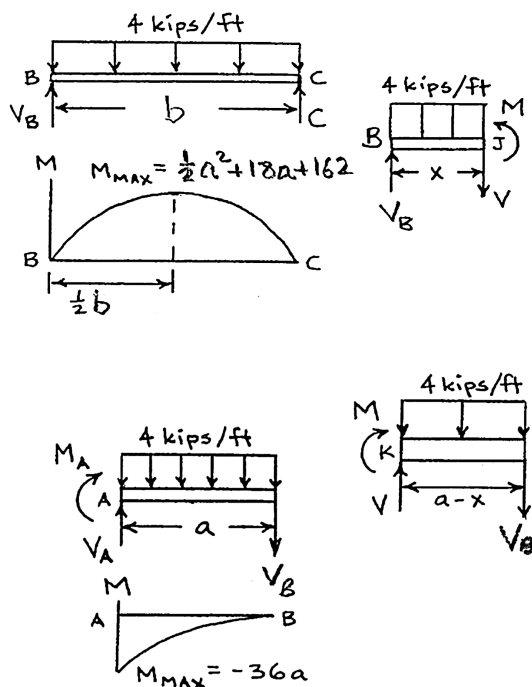
$$a = 54 \pm 50.9118 = 3.0883 \text{ ft}$$

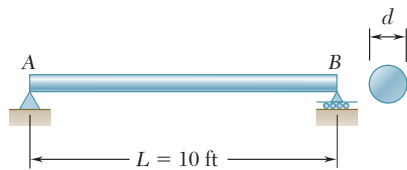
$$a = 3.09 \text{ ft} \quad \blacktriangleleft$$

(b) $|M|_{\max} = 36a = 111.179 \text{ kip} \cdot \text{ft} = 1334.15 \text{ kip} \cdot \text{in.}$

$$\sigma = \frac{|M|_{\max}}{S_x} = \frac{1334.15}{103} = 12.95 \text{ kips/in}^2$$

$$\sigma_m = 12.95 \text{ ksi} \quad \blacktriangleleft$$





PROBLEM 5.32

A solid steel rod of diameter d is supported as shown. Knowing that for steel $\gamma = 490 \text{ lb/ft}^3$, determine the smallest diameter d that can be used if the normal stress due to bending is not to exceed 4 ksi.

SOLUTION

Let W = total weight.

$$W = AL\gamma = \frac{\pi}{4}d^2L\gamma$$

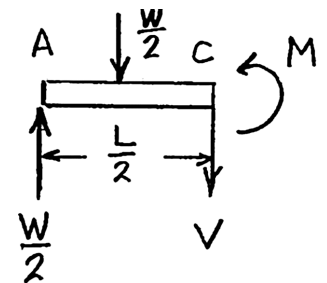
Reaction at A :

$$A = \frac{1}{2}W$$

Bending moment at center of beam:

$$+\circlearrowleft \Sigma M_C = 0: -\left(\frac{W}{2}\right)\left(\frac{L}{2}\right) + \left(\frac{W}{2}\right)\left(\frac{L}{4}\right) + M = 0$$

$$M = \frac{WL}{8} = \frac{\pi}{32}d^2L^2\gamma$$



For circular cross section, $\left(c = \frac{1}{2}d\right)$

$$I = \frac{\pi}{4}c^4, \quad S = \frac{I}{c} = \frac{\pi}{4}c^3 = \frac{\pi}{32}d^3$$

Normal stress:

$$\sigma = \frac{M}{S} = \frac{\frac{\pi}{32}d^2L^2\gamma}{\frac{\pi}{32}d^3} = \frac{L^2\gamma}{d}$$

Solving for d ,

$$d = \frac{L^2\gamma}{\sigma}$$

Data:

$$L = 10 \text{ ft} = (12)(10) = 120 \text{ in.}$$

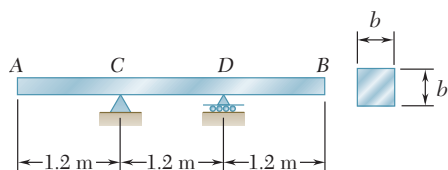
$$\gamma = 490 \text{ lb/ft}^3 = \frac{490}{12^3} = 0.28356 \text{ lb/in}^3$$

$$\sigma = 4 \text{ ksi} = 4000 \text{ lb/in}^2$$

$$d = \frac{(120)^2(0.28356)}{4000}$$

$$d = 1.021 \text{ in.} \quad \blacktriangleleft$$

PROBLEM 5.33



A solid steel bar has a square cross section of side b and is supported as shown. Knowing that for steel $\rho = 7860 \text{ kg/m}^3$, determine the dimension b for which the maximum normal stress due to bending is (a) 10 MPa, (b) 50 MPa.

SOLUTION

Weight density: $\gamma = \rho g$

Let L = total length of beam.

$$W = AL\rho g = b^2L\rho g$$

Reactions at C and D:

$$C = D = \frac{W}{2}$$

Bending moment at C:

$$+\circlearrowleft \Sigma M_C = 0: \left(\frac{L}{6}\right)\left(\frac{W}{3}\right) + M = 0$$

$$M = -\frac{WL}{18}$$

Bending moment at center of beam:

$$+\circlearrowleft \Sigma M_E = 0: \left(\frac{L}{4}\right)\left(\frac{W}{2}\right) - \left(\frac{L}{6}\right)\left(\frac{W}{2}\right) + M = 0 \quad M = -\frac{WL}{24}$$

$$\max |M| = \frac{WL}{18} = \frac{b^2L^2\rho g}{18}$$

For a square section,

$$S = \frac{1}{6} b^3$$

Normal stress:

$$\sigma = \frac{|M|}{S} = \frac{b^2L^2\rho g/18}{b^3/6} = \frac{L^2\rho g}{3b}$$

Solve for b :

$$b = \frac{L^2\rho g}{3\sigma}$$

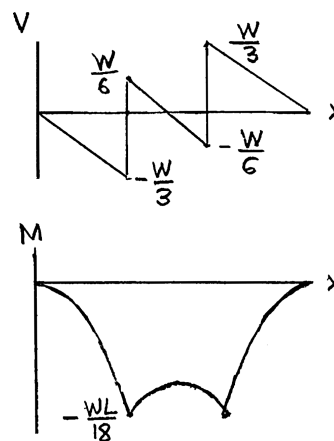
Data: $L = 3.6 \text{ m}$ $\rho = 7860 \text{ kg/m}^3$ $g = 9.81 \text{ m/s}^2$ (a) $\sigma = 10 \times 10^6 \text{ Pa}$ (b) $\sigma = 50 \times 10^6 \text{ Pa}$

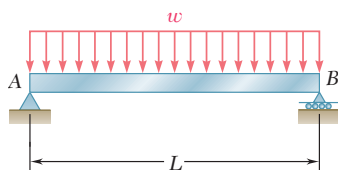
$$(a) \quad b = \frac{(3.6)^2(7860)(9.81)}{(3)(10 \times 10^6)} = 33.3 \times 10^{-3} \text{ m}$$

$$b = 33.3 \text{ mm} \quad \blacktriangleleft$$

$$(b) \quad b = \frac{(3.6)^2(7860)(9.81)}{(3)(50 \times 10^6)} = 6.66 \times 10^{-3} \text{ m}$$

$$b = 6.66 \text{ mm} \quad \blacktriangleleft$$





PROBLEM 5.34

Using the method of Sec. 5.2, solve Prob. 5.1a.

PROBLEM 5.1 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

$$+\circlearrowleft \Sigma M_B = 0: -AL + wL \cdot \frac{L}{2} = 0 \quad A = \frac{wL}{2}$$

$$+\circlearrowleft \Sigma M_A = 0: BL - wL \cdot \frac{L}{2} = 0 \quad B = \frac{wL}{2}$$

$$\frac{dV}{dx} = -w$$

$$V - V_A = -\int_0^x w dx = -wx$$

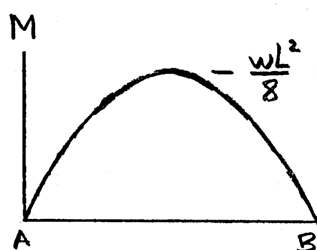
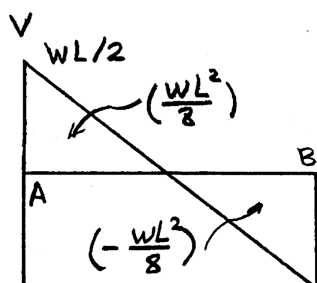
$$V = V_A - wx = A - wx \quad V = \frac{wL}{2} - wx \quad \blacktriangleleft$$

$$\frac{dM}{dx} = V$$

$$M - M_A = \int_0^x V dx = \int_0^x \left(\frac{wL}{2} - wx \right) dx$$

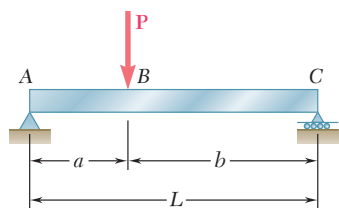
$$= \frac{wLx}{2} - \frac{wx^2}{2}$$

$$M = M_A + \frac{wLx}{2} - \frac{wx^2}{2} \quad M = \frac{w}{2}(Lx - x^2) \quad \blacktriangleleft$$



Maximum M occurs at $x = \frac{1}{2}$, where

$$V = \frac{dM}{dx} = 0 \quad |M|_{\max} = \frac{wL^2}{8} \quad \blacktriangleleft$$



PROBLEM 5.35

Using the method of Sec. 5.2, solve Prob. 5.2a.

PROBLEM 5.2 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

$$+\circlearrowleft \Sigma M_C = 0: LA - bP = 0 \quad A = \frac{Pb}{L}$$

$$+\circlearrowleft \Sigma M_A = 0: LC - aP = 0 \quad C = \frac{Pa}{L}$$

At A^+

$$V = A = \frac{Pb}{L} \quad M = 0$$

A to B^- :

$$0 < x < a$$

$$w = 0 \quad \int_0^x w dx = 0$$

$$V - V_A = 0$$

$$M_B - M_A = \int_0^a V dx = \int_0^a \frac{Pb}{L} dx = \frac{Pba}{L}$$

At B^+

$$V = A - P = \frac{Pb}{L} - P = -\frac{Pa}{L}$$

B^+ to C:

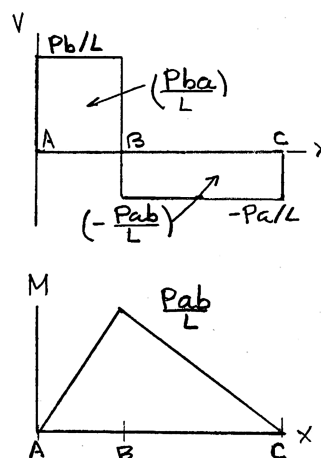
$$a < x < L$$

$$w = 0 \quad \int_a^x w dx = 0$$

$$V_C - V_B = 0$$

$$M_C - M_B = \int_a^L V dx = -\frac{Pa}{L}(L - a) = -\frac{Pab}{L}$$

$$M_C = M_B - \frac{Pab}{L} = \frac{Pba}{L} - \frac{Pab}{L} = 0$$



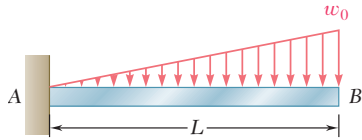
$$V = \frac{Pb}{L} \quad \blacktriangleleft$$

$$M_B = \frac{Pba}{L} \quad \blacktriangleleft$$

$$V = -\frac{Pa}{L} \quad \blacktriangleleft$$

$$|M|_{\max} = \frac{Pab}{L} \quad \blacktriangleleft$$

PROBLEM 5.36

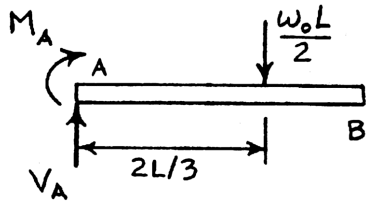


Using the method of Sec. 5.2, solve Prob. 5.3a.

PROBLEM 5.3 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

Free body diagram for determining reactions.

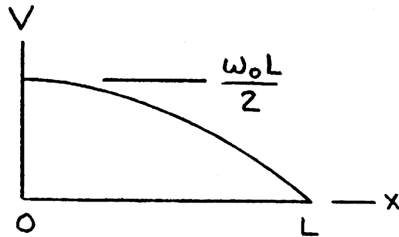


Reactions:

$$+\uparrow \Sigma F_y = 0: V_A - \frac{w_0 L}{2} = 0 \quad V_A = \frac{w_0 L}{2}$$

$$+\curvearrowright \Sigma M_A = 0: -M_A - \left(\frac{w_0 L}{2} \right) \left(\frac{2L}{3} \right) = 0$$

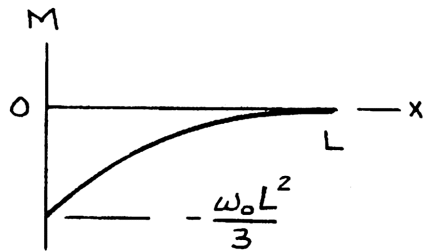
$$M_A = -\frac{w_0 L^2}{3}$$



$$w = w_0 \frac{x}{L}, \quad V_A = \frac{w_0 L}{2}, \quad M_A = -\frac{w_0 L^2}{3}$$

$$\frac{dV}{dx} = -w = -\frac{w_0 x}{L}$$

$$V - V_A = -\int_0^x \frac{w_0 x}{L} dx = -\frac{w_0 x^2}{2L}$$



$$V = \frac{w_0 L}{2} - \frac{w_0 x^2}{2L} \quad \blacktriangleleft$$

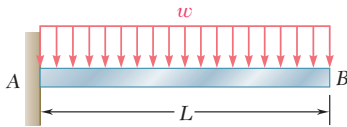
$$\frac{dM}{dx} = V = \frac{w_0 L}{2} - \frac{w_0 x^2}{2L}$$

$$M - M_A = \int_0^x V dx = \int_0^x \left(\frac{w_0 L}{2} - \frac{w_0 x^2}{2L} \right) dx$$

$$= \frac{w_0 L}{2} x - \frac{w_0 x^3}{6L}$$

$$M = -\frac{w_0 L^2}{3} + \frac{w_0 L}{2} x - \frac{w_0 x^3}{6L} \quad \blacktriangleleft$$

PROBLEM 5.37



Using the method of Sec. 5.2, solve Prob. 5.4a.

PROBLEM 5.4 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION

$$+\uparrow \Sigma F_y = 0: V_A - wL = 0 \quad V_A = wL$$

$$+\curvearrowright \Sigma M_A = 0: -M - (wL)\left(\frac{L}{2}\right) = 0 \quad M_A = -\frac{wL^2}{2}$$

$$\frac{dV}{dx} = -w$$

$$V - V_A = -\int_0^x w dx = -wx$$

$$V = wL - wx \quad \blacktriangleleft$$

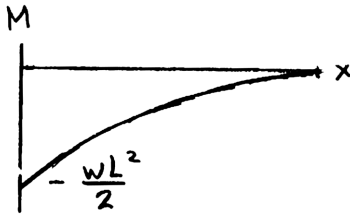
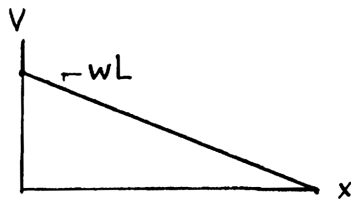
$$\frac{dM}{dx} = V = wL - wx$$

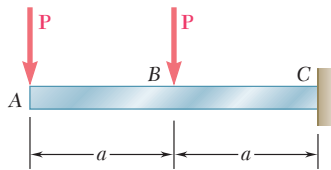
$$M - M_A = -\int_0^x (wL - wx) dx = wLx - \frac{wx^2}{2}$$

$$M = -\frac{wL^2}{2} + wLx - \frac{wx^2}{2} \quad \blacktriangleleft$$

$$|V|_{\max} = wL$$

$$|M|_{\max} = \frac{wL^2}{2}$$



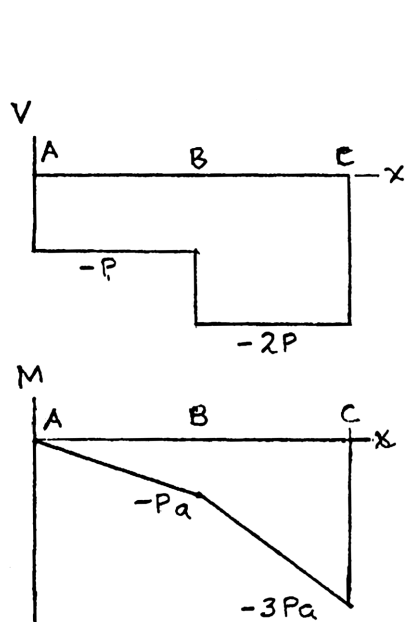


PROBLEM 5.38

Using the method of Sec. 5.2, solve Prob. 5.5a.

PROBLEM 5.5 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.

SOLUTION



At A^+ : $V_A = -P$

Over AB : $\frac{dV}{dx} = -w = 0$

$$\frac{dM}{dx} = V = V_A = -P \quad \blacktriangleleft$$

$$M = -Px + C$$

$$M = 0 \quad \text{at } x = 0 \quad C_1 = 0$$

$$M = -Px \quad \blacktriangleleft$$

At point B : $x = a \quad M = -Pa$

At point B^+ : $V = -P - P = -2P$

Over BC : $\frac{dV}{dx} = -w = 0$

$$\frac{dM}{dx} = V = -2P \quad \blacktriangleleft$$

$$M = -2Px + C_2$$

At B : $x = a \quad M = -Pa$

$$-Pa = -2Pa + C_2 \quad C_2 = Pa$$

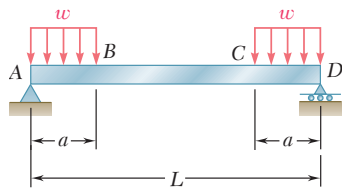
$$M = -2Px + Pa \quad \blacktriangleleft$$

At C : $x = 2a \quad M = -3Pa$

PROBLEM 5.39

Using the method of Sec. 5.2, solve Prob. 5.6a.

PROBLEM 5.6 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the equations of the shear and bending-moment curves.



SOLUTION

Reactions: $A = D = wa$

A to B: $0 < x < a$ $w = w$

$$V_A = A = wa, \quad M_A = 0$$

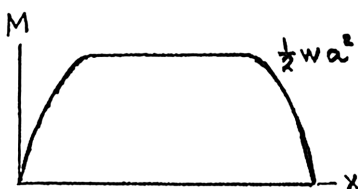
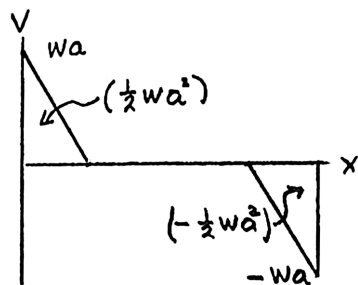
$$V - V_A = -\int_0^x w dx = -wx$$

$$V = w(a - x) \quad \blacktriangleleft$$

$$\frac{dM}{dx} = V = wa - wx$$

$$M - M_A = \int_0^x V dx = \int_0^x (wa - wx) dx$$

$$M = wax - \frac{1}{2}wx^2 \quad \blacktriangleleft$$



$$V_B = 0 \quad M_B = \frac{1}{2}wa^2$$

B to C: $a < x < L - a$

$$V = 0 \quad \blacktriangleleft$$

$$\frac{dM}{dx} = V = 0$$

$$M - M_B = \int_a^x V dx = 0$$

$$M = M_B$$

$$M = \frac{1}{2}wa^2 \quad \blacktriangleleft$$

PROBLEM 5.39 (Continued)

$$L - a < x < L$$

$$C \text{ to } D: \quad V - V_C = -\int_{L-a}^x w \, dx = -w[x - (L - a)]$$

$$V = w[L - x - a] \quad \blacktriangleleft$$

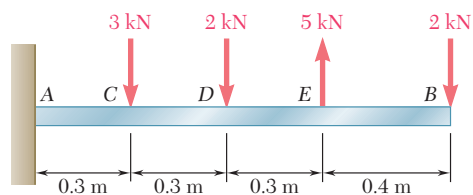
$$M - M_C = \int_{L-a}^x V \, dx = \int_{L-a}^x -w[x - (L - a)] \, dx$$

$$= -w \left[\frac{x^2}{2} - (L - a)x \right] \Big|_{L-a}^x$$

$$= -w \left[\frac{x^2}{2} - (L - a)x - \frac{(L - a)^2}{2} + (L - a)^2 \right]$$

$$= -w \left[\frac{x^2}{2} - (L - a)x + \frac{(L - a)^2}{2} \right]$$

$$M = \frac{1}{2}wa^2 - w \left[\frac{x^2}{2} - (L - a)x + \frac{(L - a)^2}{2} \right] \quad \blacktriangleleft$$



PROBLEM 5.40

Using the method of Sec. 5.2, solve Prob. 5.7.

PROBLEM 5.7 Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Free body diagram for determining reactions.

Reactions:

$$+\uparrow \Sigma F_y = 0: V_A - 3 \text{ kN} - 2 \text{ kN} + 5 \text{ kN} + 2 \text{ kN} = 0$$

$$V_A = 2 \text{ kN}$$

$$+\circlearrowleft \Sigma M_A = 0: M_A - (3 \text{ kN})(0.3 \text{ m}) - (2 \text{ kN})(0.6 \text{ m}) + (5 \text{ kN})(0.9 \text{ m}) - (2 \text{ kN})(1.3 \text{ m}) = 0$$

$$M_A = 0.2 \text{ kN} \cdot \text{m}$$

Between concentrated loads and the vertical reaction, the slope of the shear diagram is ϕ , i.e., the shear is constant. Thus, the area under the shear diagram is equal to the change in bending moment.

A to C:

$$V = 2 \text{ kN} \quad M_C - M_A = +0.6 \quad M_C = +0.4 \text{ kN}$$

C to D:

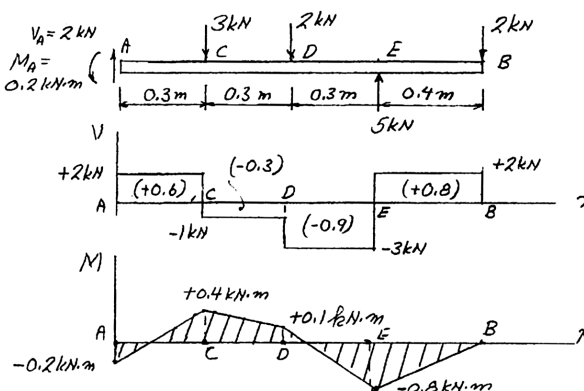
$$V = -1 \text{ kN} \quad M_D - M_C = -0.3 \quad M_D = +0.1 \text{ kN} \cdot \text{m}$$

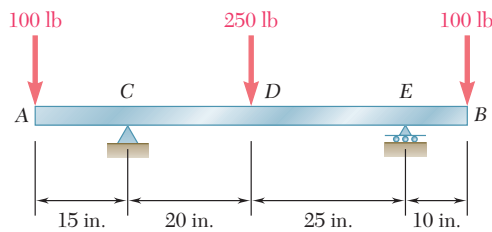
D to E:

$$V = -3 \text{ kN} \quad M_E - M_D = -0.9 \quad M_E = -0.8 \text{ kN} \cdot \text{m}$$

E to B:

$$V = +2 \text{ kN} \quad M_B - M_E = +0.8 \quad M_B = 0 \quad (\text{Checks})$$





PROBLEM 5.41

Using the method of Sec. 5.2, solve Prob. 5.8.

PROBLEM 5.8 Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Free body diagram for determining reactions.

Reactions:

$$+\uparrow \Sigma F_Y = 0: V_C + V_E - 100 \text{ lb} - 250 \text{ lb} - 100 \text{ lb} = 0$$

$$V_C = V_E = 450 \text{ lb}$$

$$+\circlearrowleft \Sigma M_C = 0: V_E(45 \text{ in.}) + (100 \text{ lb})(15 \text{ in.}) - (250 \text{ lb})(20 \text{ in.}) - (100 \text{ lb})(55 \text{ in.}) = 0$$

$$V_E = 200 \text{ lb}$$

$$\therefore V_C = 250 \text{ lb}$$

Between concentrated loads and the vertical reaction, the slope of the shear diagram is ϕ , i.e., the shear is constant. Thus, the area under the shear diagram is equal to the change in bending moment.

A to C:

$$V = -100 \text{ lb}, \quad M_C - M_A = -1500, \quad M_C = -1500 \text{ lb} \cdot \text{in.}$$

C to D:

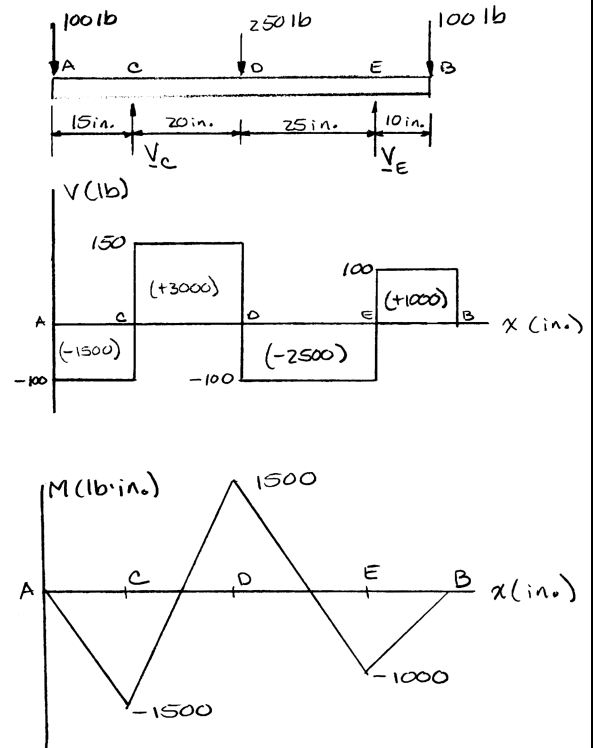
$$V = 150 \text{ lb}, \quad M_D - M_C = +3000, \quad M_D = +1500 \text{ lb} \cdot \text{in.}$$

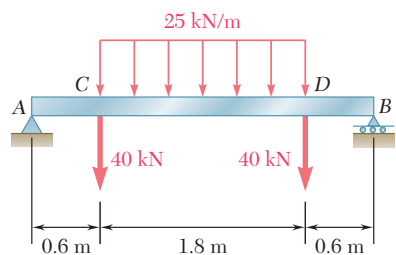
D to E:

$$V = -100 \text{ lb}, \quad M_E - M_D = -2500, \quad M_E = -1000 \text{ lb} \cdot \text{in.}$$

E to B:

$$V = 100 \text{ lb}, \quad M_B - M_E = +1000, \quad M_B = 0 \text{ (Checks)}$$





PROBLEM 5.42

Using the method of Sec. 5.2, solve Prob. 5.9.

PROBLEM 5.9 Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION

Free body diagram to determine reactions:

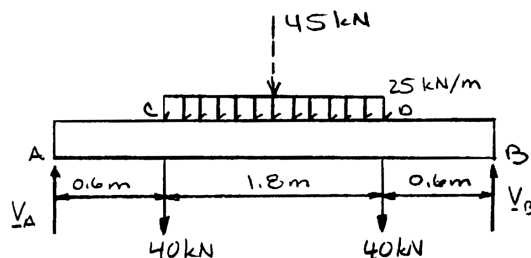
$$+\circlearrowleft \Sigma M_A = 0:$$

$$V_B(3.0 \text{ m}) - 45 \text{ kN}(1.5 \text{ m}) - (40 \text{ kN})(0.6 \text{ m}) - (40 \text{ kN})(2.4 \text{ m}) = 0$$

$$V_B = 62.5 \text{ kN}$$

$$+\uparrow \Sigma F_y = 0: V_A - 40 \text{ kN} - 45 \text{ kN} - 40 \text{ kN} + 62.5 \text{ kN} = 0$$

$$V_A = 62.5 \text{ kN}$$



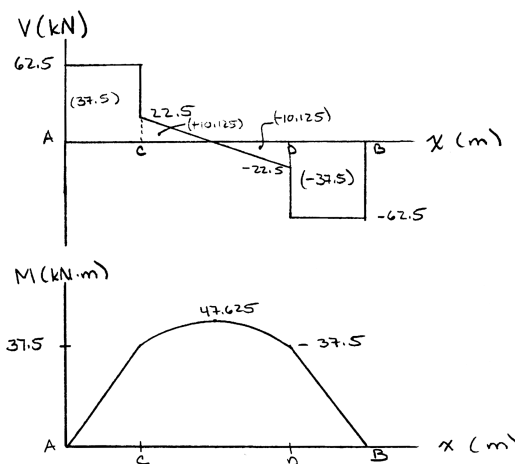
Change in bending moment is equal to area under shear diagram.

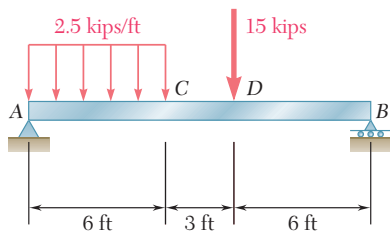
$$A \text{ to } C: (62.5 \text{ kN})(0.6 \text{ m}) = 37.5 \text{ kN} \cdot \text{m}$$

$$C \text{ to } E: \frac{1}{2}(0.9 \text{ m})(22.5 \text{ kN}) = 10.125 \text{ kN} \cdot \text{m}$$

$$E \text{ to } D: \frac{1}{2}(0.9 \text{ m})(-22.5 \text{ kN}) = -10.125 \text{ kN} \cdot \text{m}$$

$$D \text{ to } B: (-62.5 \text{ kN})(0.6 \text{ m}) = -37.5 \text{ kN} \cdot \text{m}$$



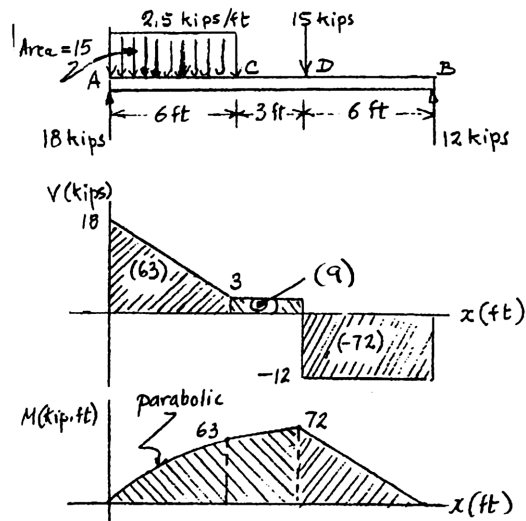


PROBLEM 5.43

Using the method of Sec. 5.2, solve Prob. 5.10.

PROBLEM 5.10 Draw the shear and bending-moment diagrams for the beam and loading shown, and determine the maximum absolute value (a) of the shear, (b) of the bending moment.

SOLUTION



Reactions at supports *A* and *B*:

$$+\circlearrowleft \sum M_B = 0: -15(R_A) + (12)(6)(2.5) + (6)(15) = 0$$

$$R_A = 18 \text{ kips}$$

$$+\circlearrowleft \sum M_A = 0: 15R_B - (3)(6)(2.5) - (9)(15) = 0$$

$$R_B = 12 \text{ kips}$$

Areas under shear diagram:

$$A \text{ to } C: (6)(3) + \frac{1}{2}(6)(15) = 63 \text{ kip} \cdot \text{ft}$$

$$C \text{ to } D: (3)(3) = 9 \text{ kip} \cdot \text{ft}$$

$$D \text{ to } B: (6)(-12) = -72 \text{ kip} \cdot \text{ft}$$

Bending moments:

$$M_A = 0$$

$$M_C = 0 + 63 = 63 \text{ kip} \cdot \text{ft}$$

$$M_D = 63 + 9 = 72 \text{ kip} \cdot \text{ft}$$

$$M_B = 72 - 72 = 0$$